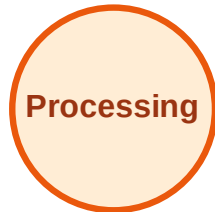


# BFS Traversal

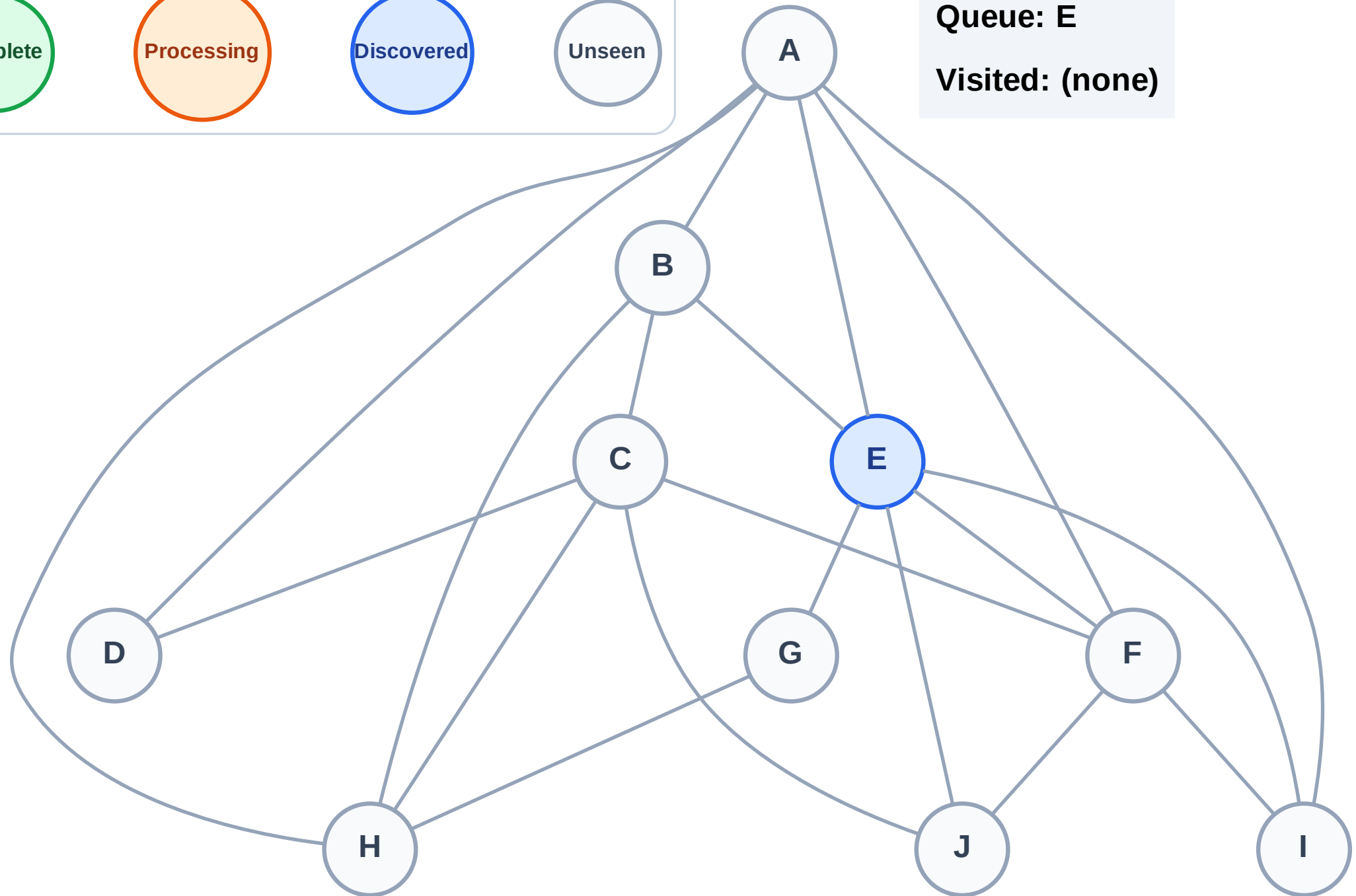
Step 1: Start at E

## Legend



Queue: E

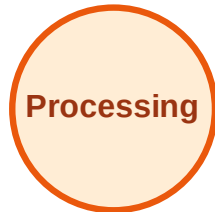
Visited: (none)



# BFS Traversal

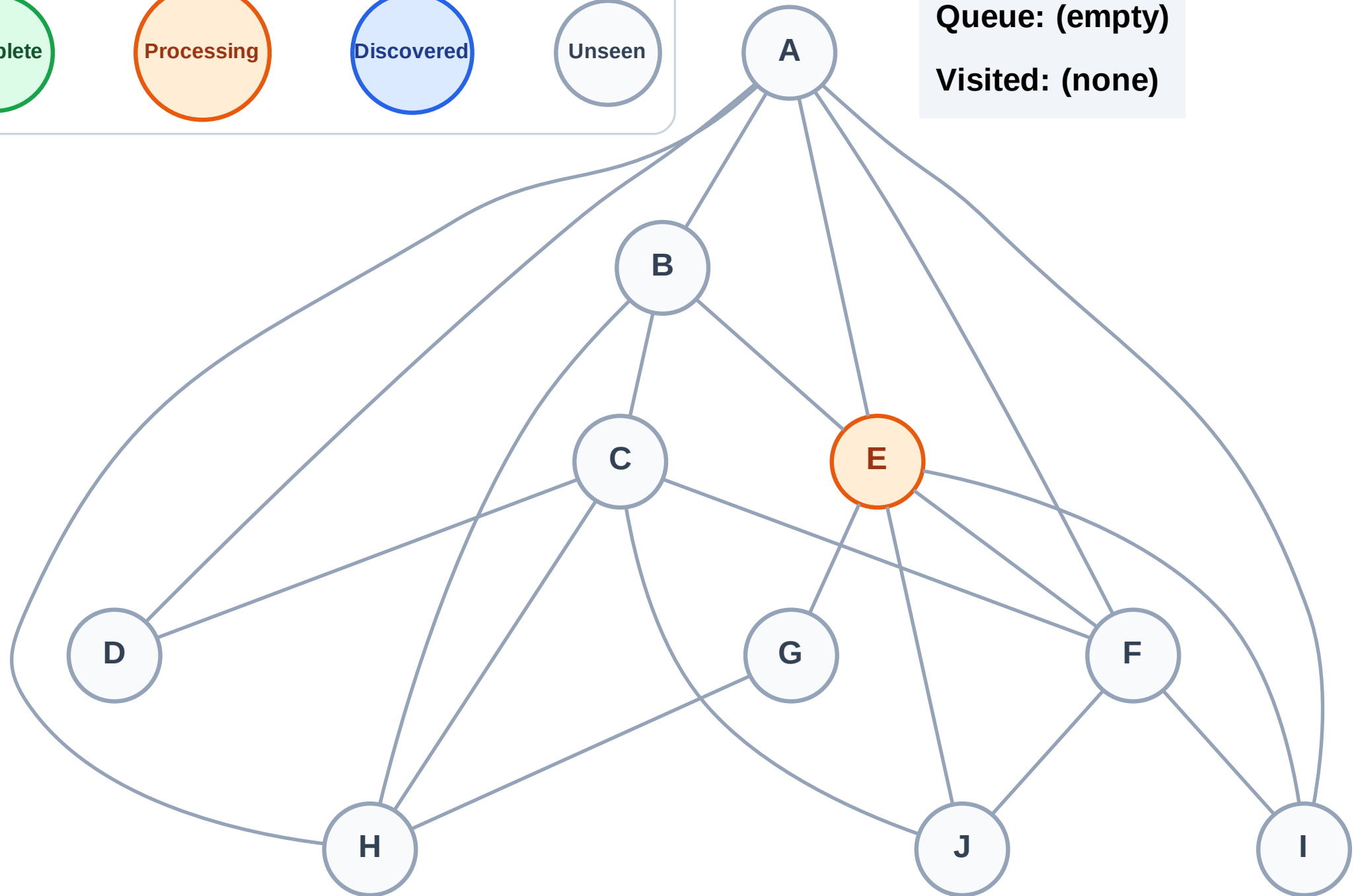
Step 2: Process node E

## Legend



Queue: (empty)

Visited: (none)



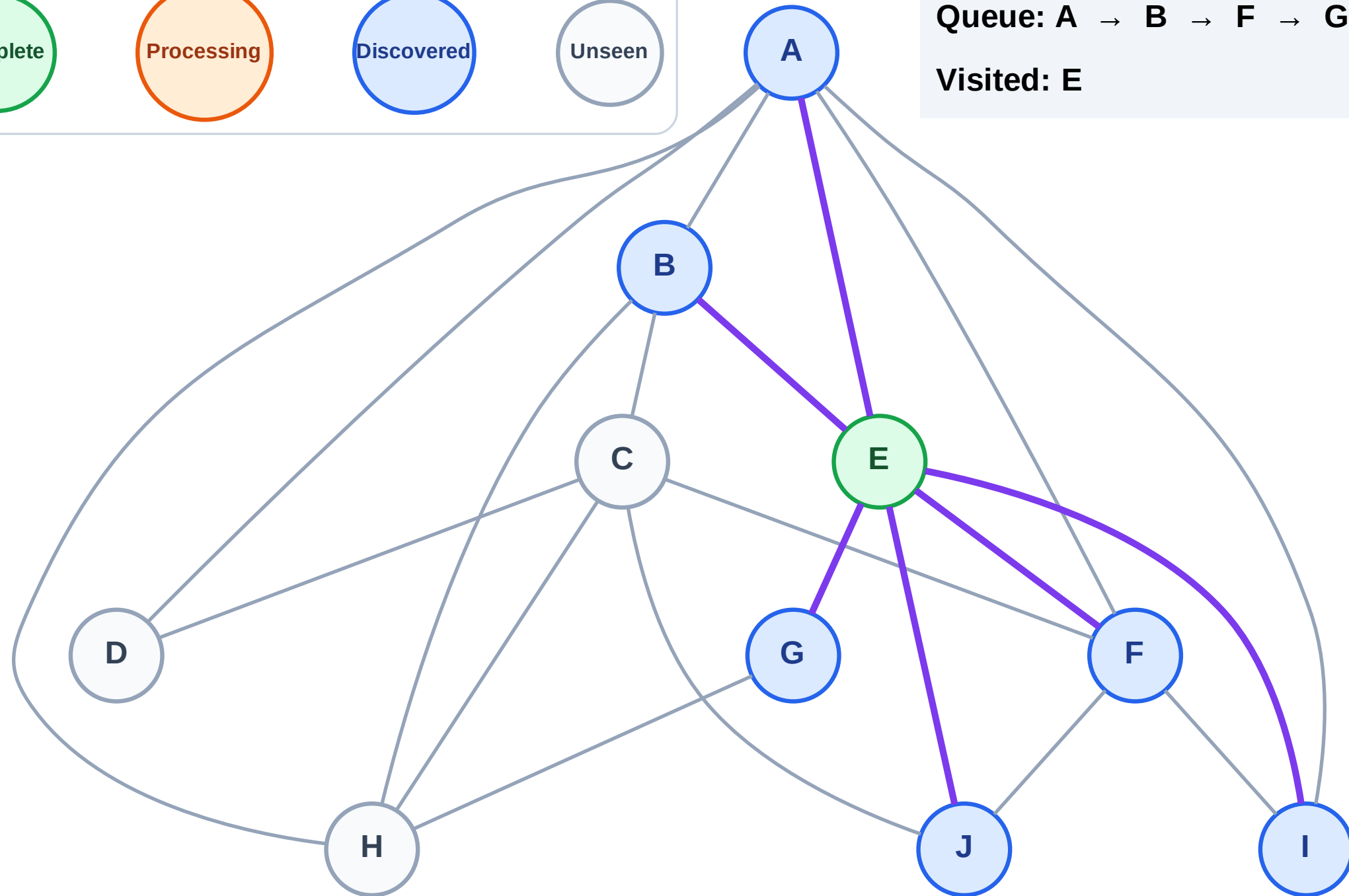
# BFS Traversal

Step 3: Discover A, B, F, G, I, J from E

## Legend



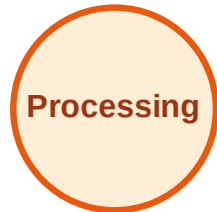
Queue: A → B → F → G → I → J  
Visited: E



# BFS Traversal

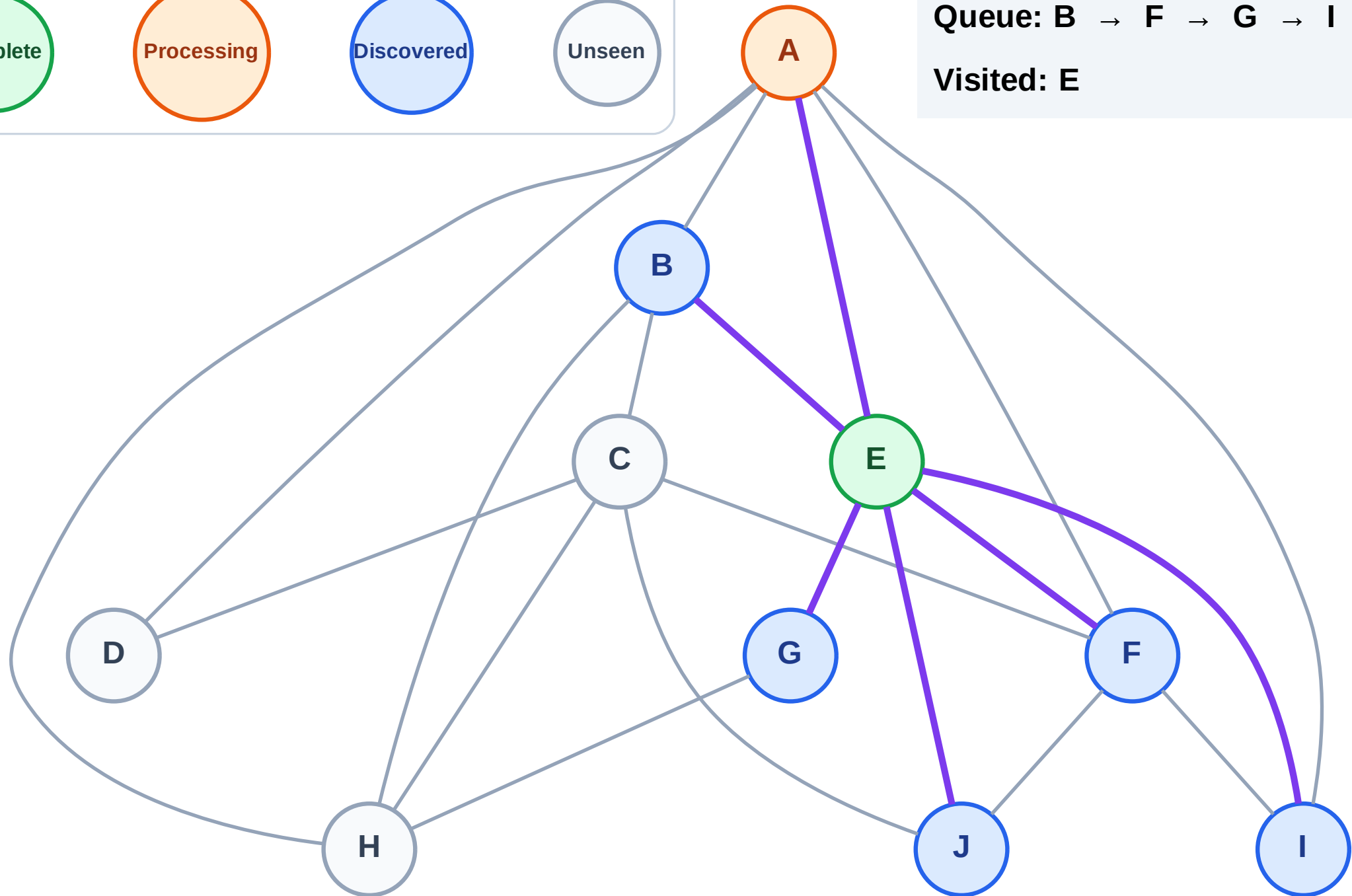
Step 4: Process node A

## Legend



Queue: B → F → G → I → J

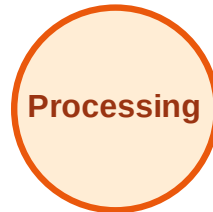
Visited: E



# BFS Traversal

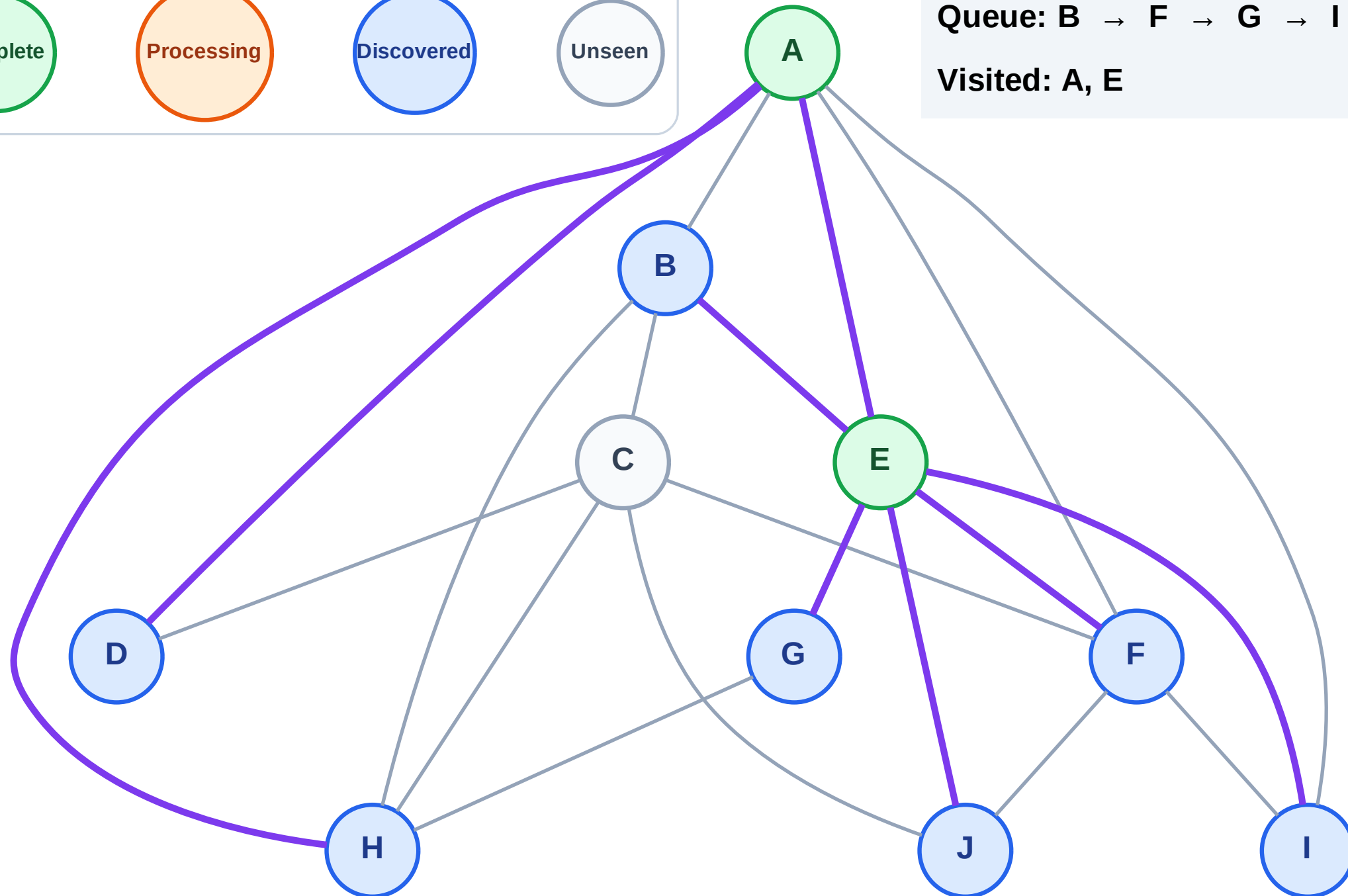
Step 5: Discover D, H from A

## Legend



Queue: B → F → G → I → J → D → H

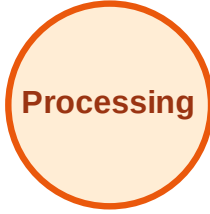
Visited: A, E



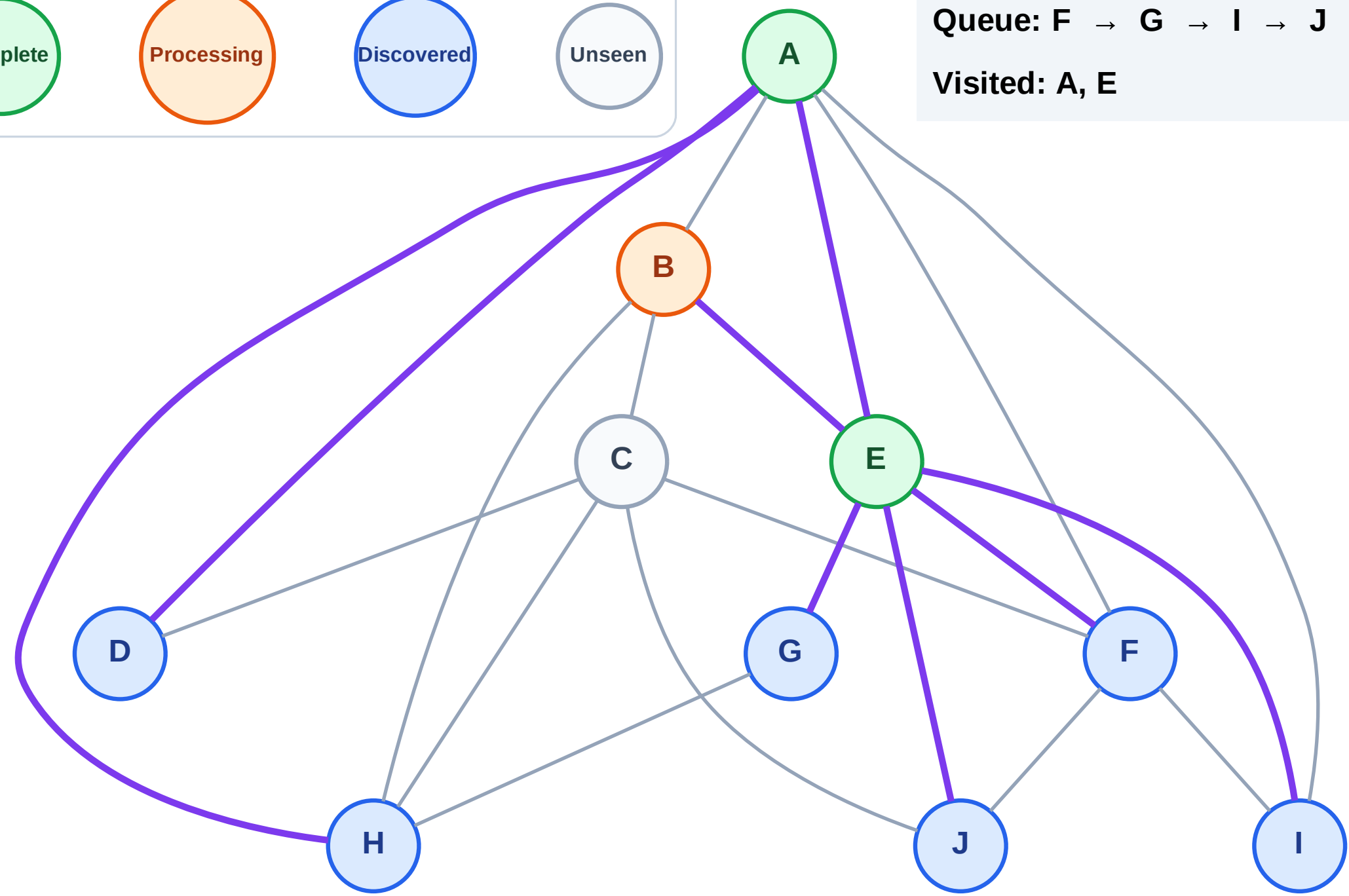
# BFS Traversal

Step 6: Process node B

## Legend



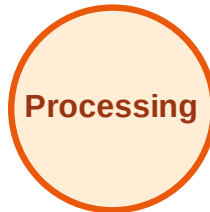
Queue: F → G → I → J → D → H  
Visited: A, E



# BFS Traversal

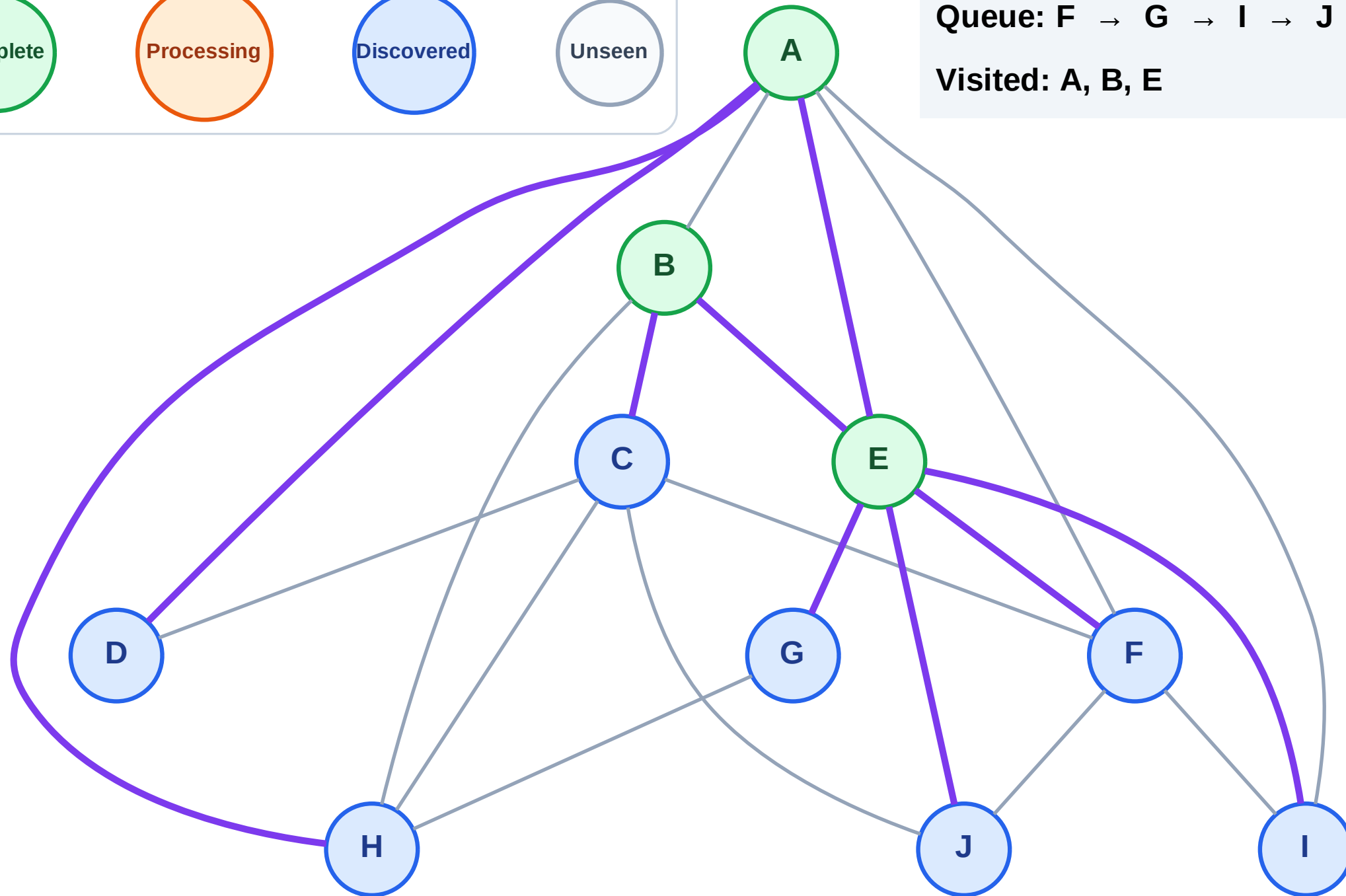
Step 7: Discover C from B

## Legend



Queue: F → G → I → J → D → H → C

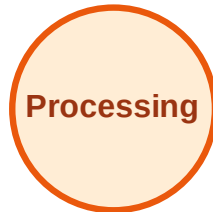
Visited: A, B, E



# BFS Traversal

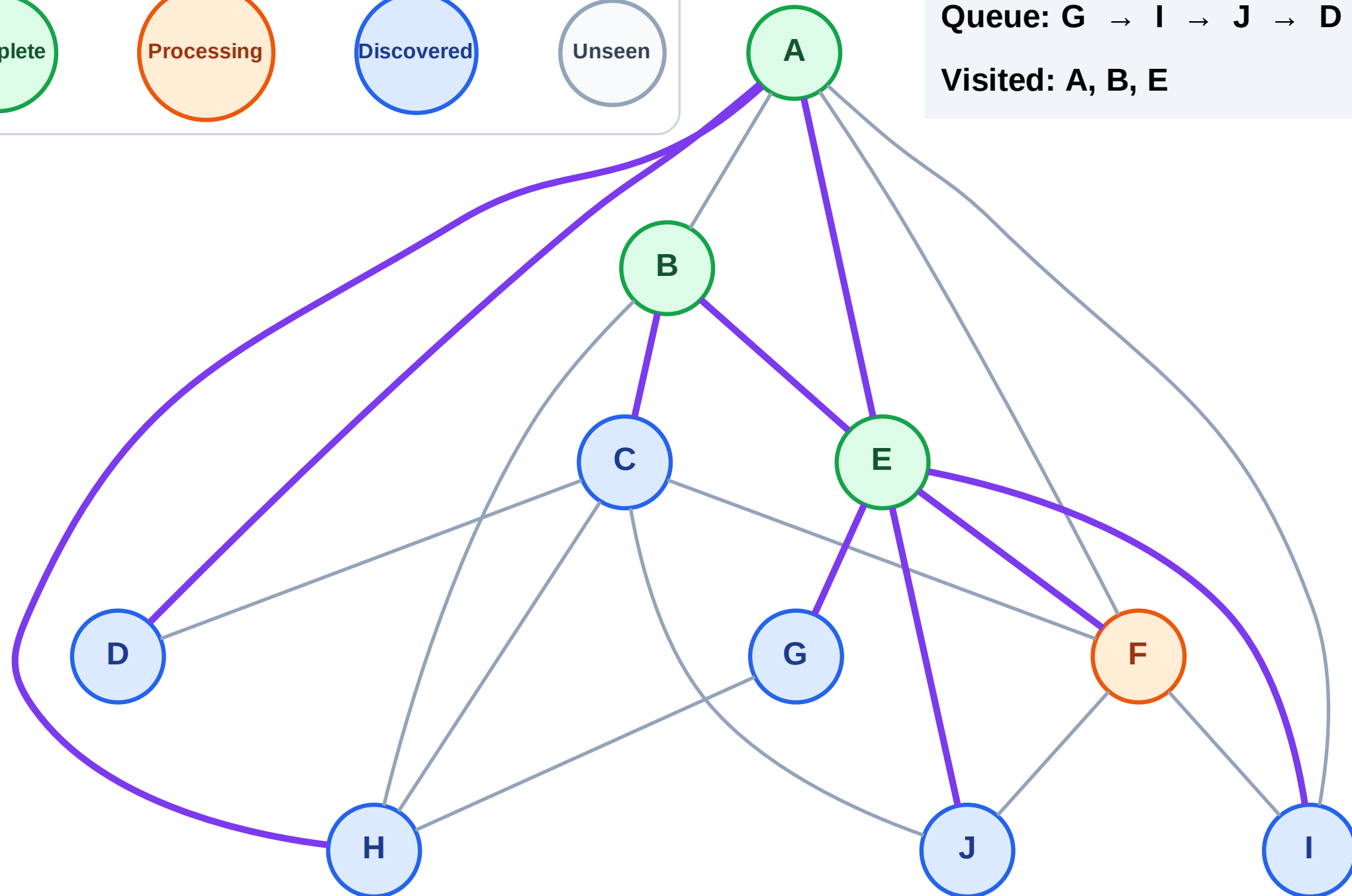
Step 8: Process node F

## Legend



Queue: G → I → J → D → H → C

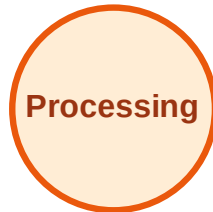
Visited: A, B, E



# BFS Traversal

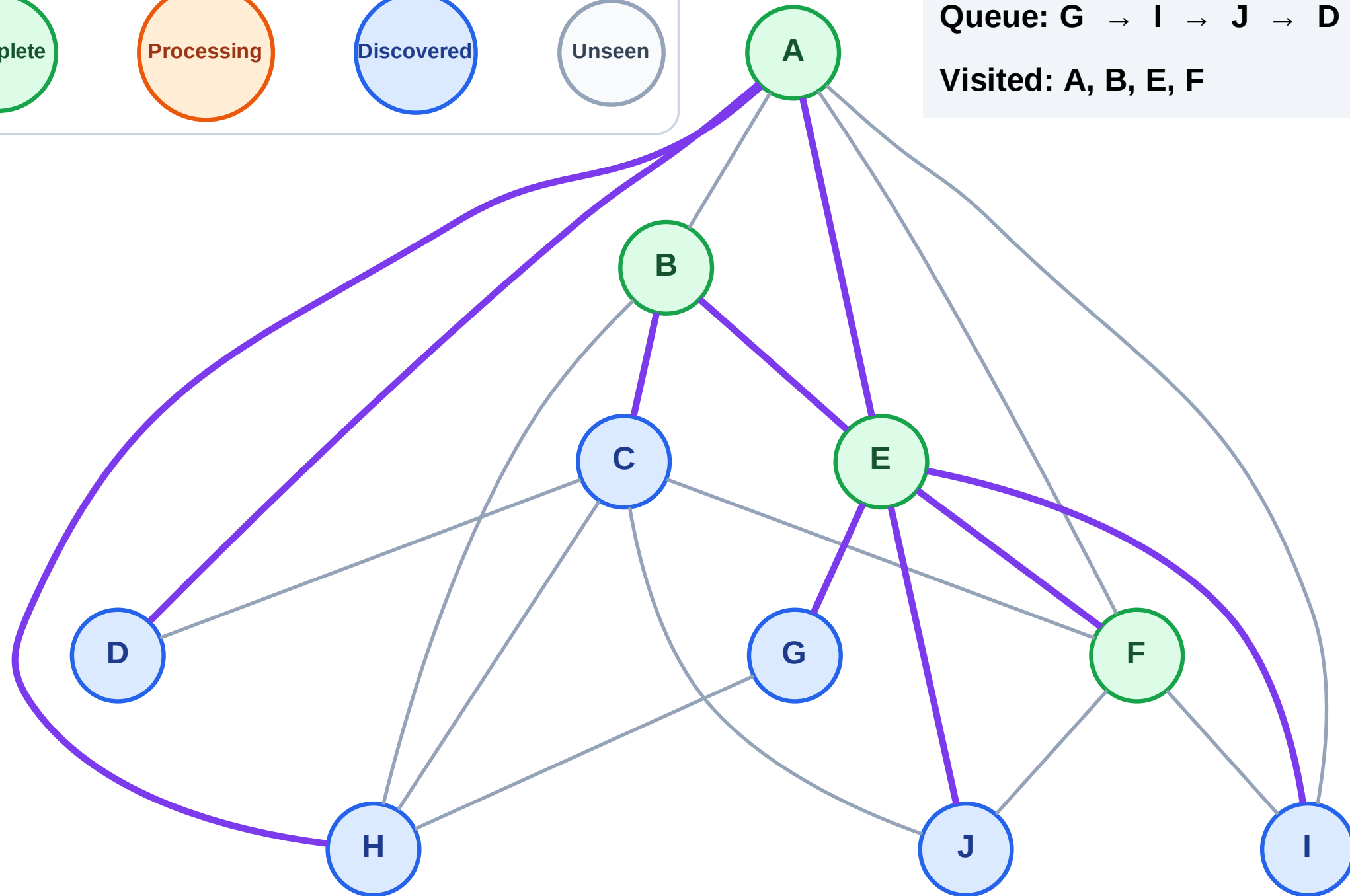
Step 9: No new neighbors from F

## Legend



Queue: G → I → J → D → H → C

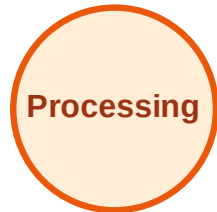
Visited: A, B, E, F



# BFS Traversal

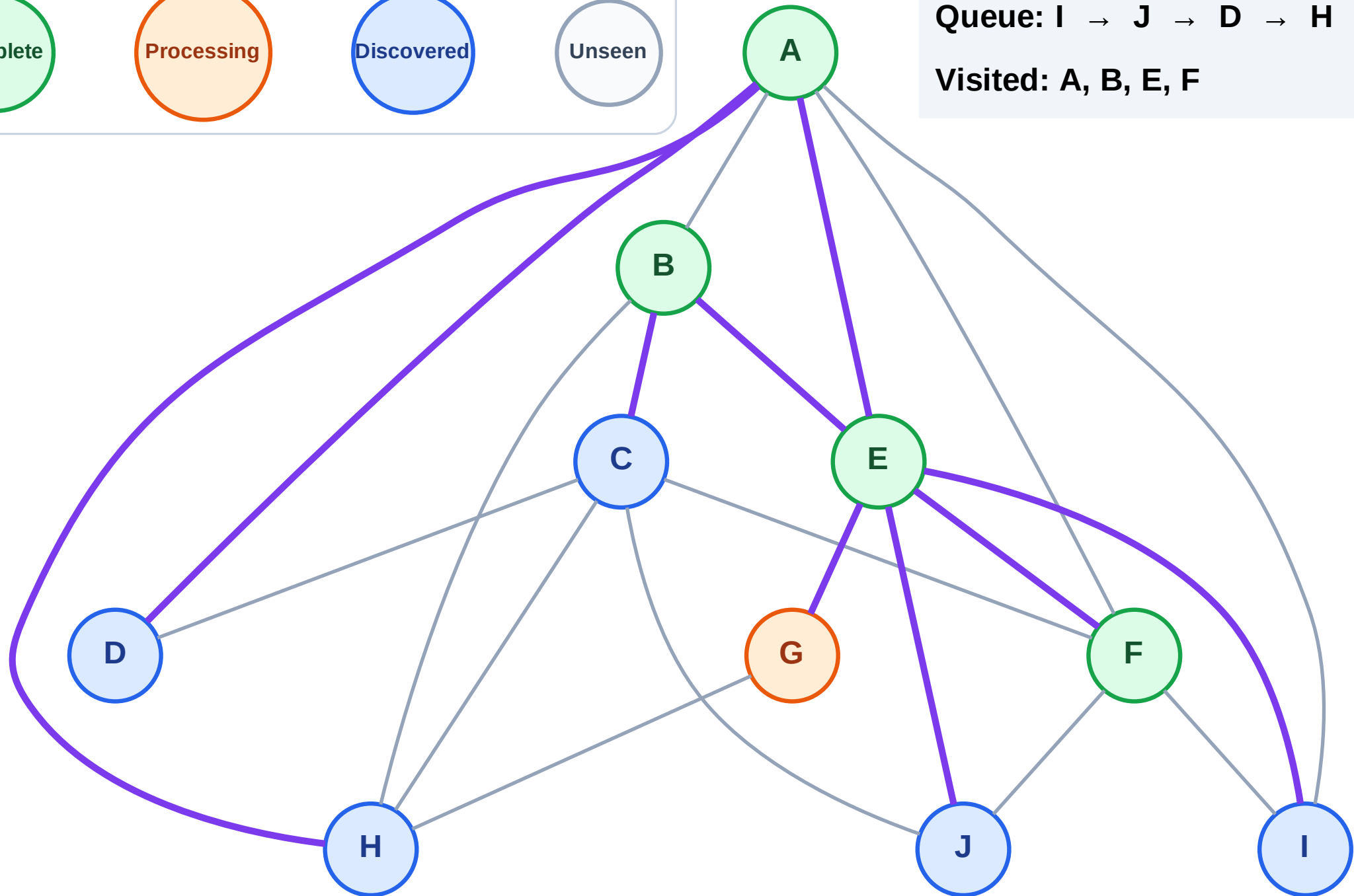
Step 10: Process node G

## Legend



Queue: I → J → D → H → C

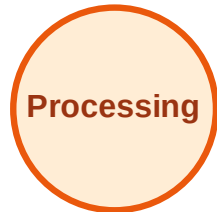
Visited: A, B, E, F



# BFS Traversal

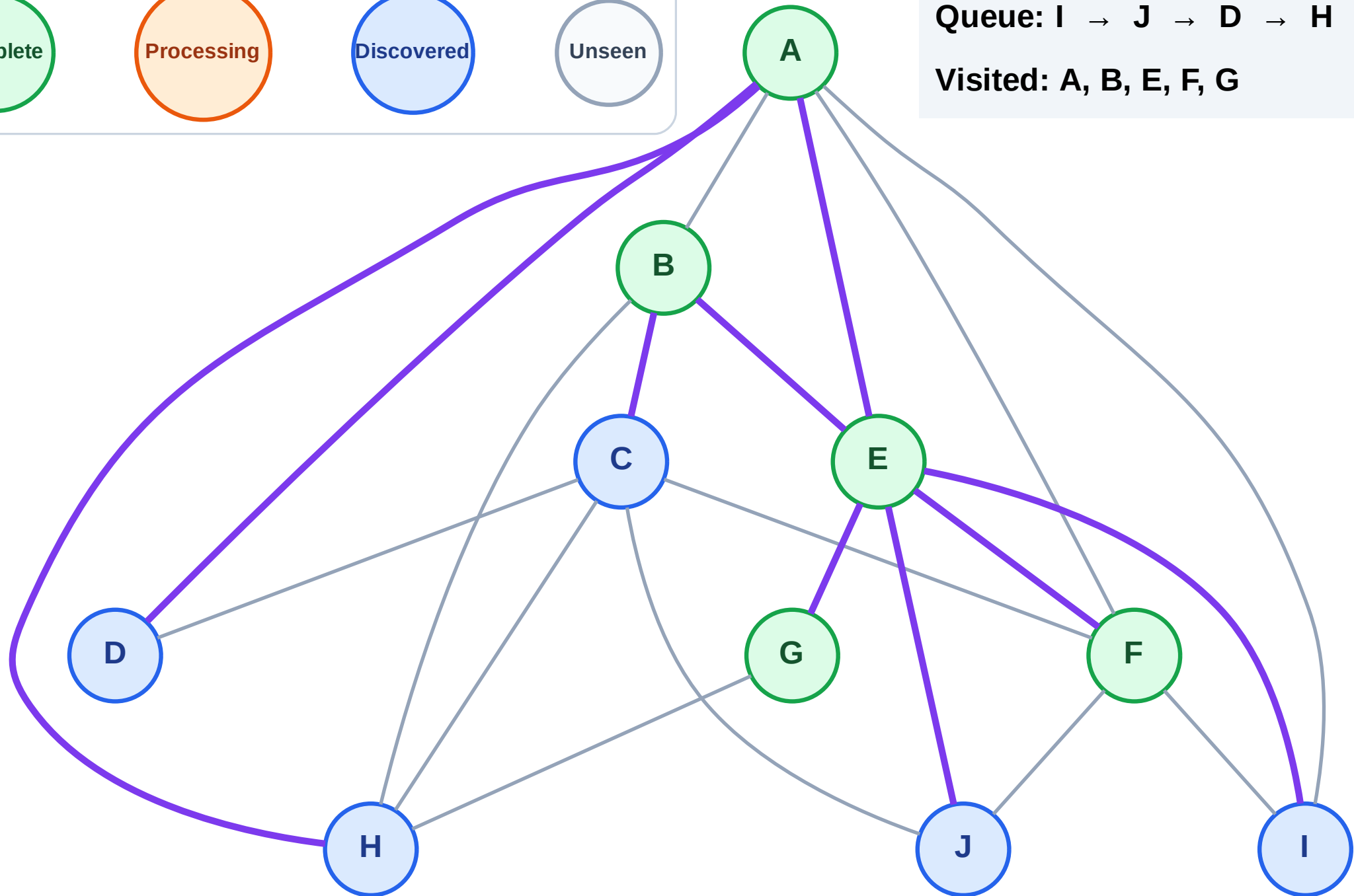
Step 11: No new neighbors from G

## Legend



Queue: I → J → D → H → C

Visited: A, B, E, F, G



# BFS Traversal

Step 12: Process node I

## Legend

Complete

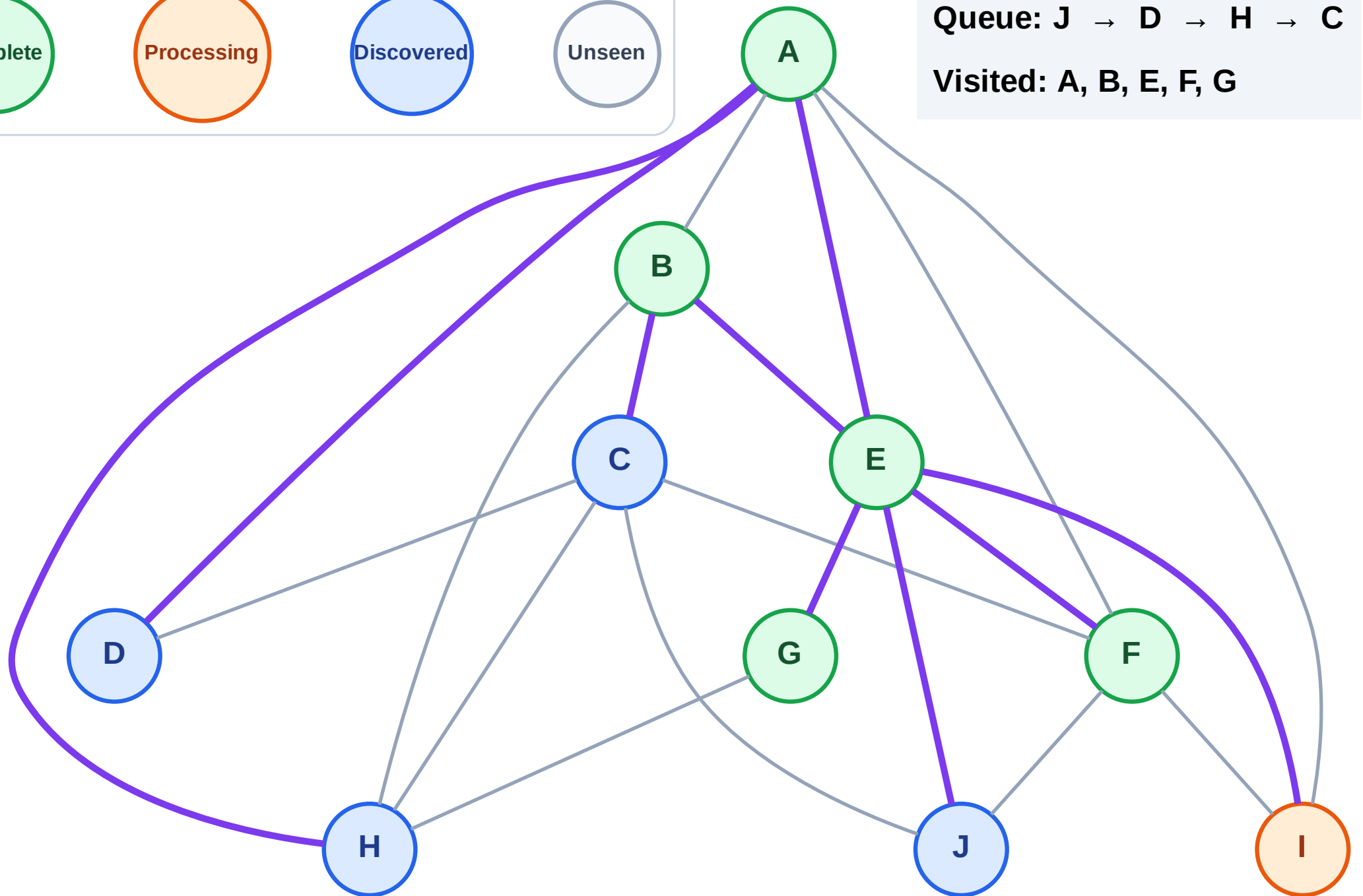
Processing

Discovered

Unseen

Queue: J → D → H → C

Visited: A, B, E, F, G



# BFS Traversal

Step 13: No new neighbors from I

## Legend

Complete

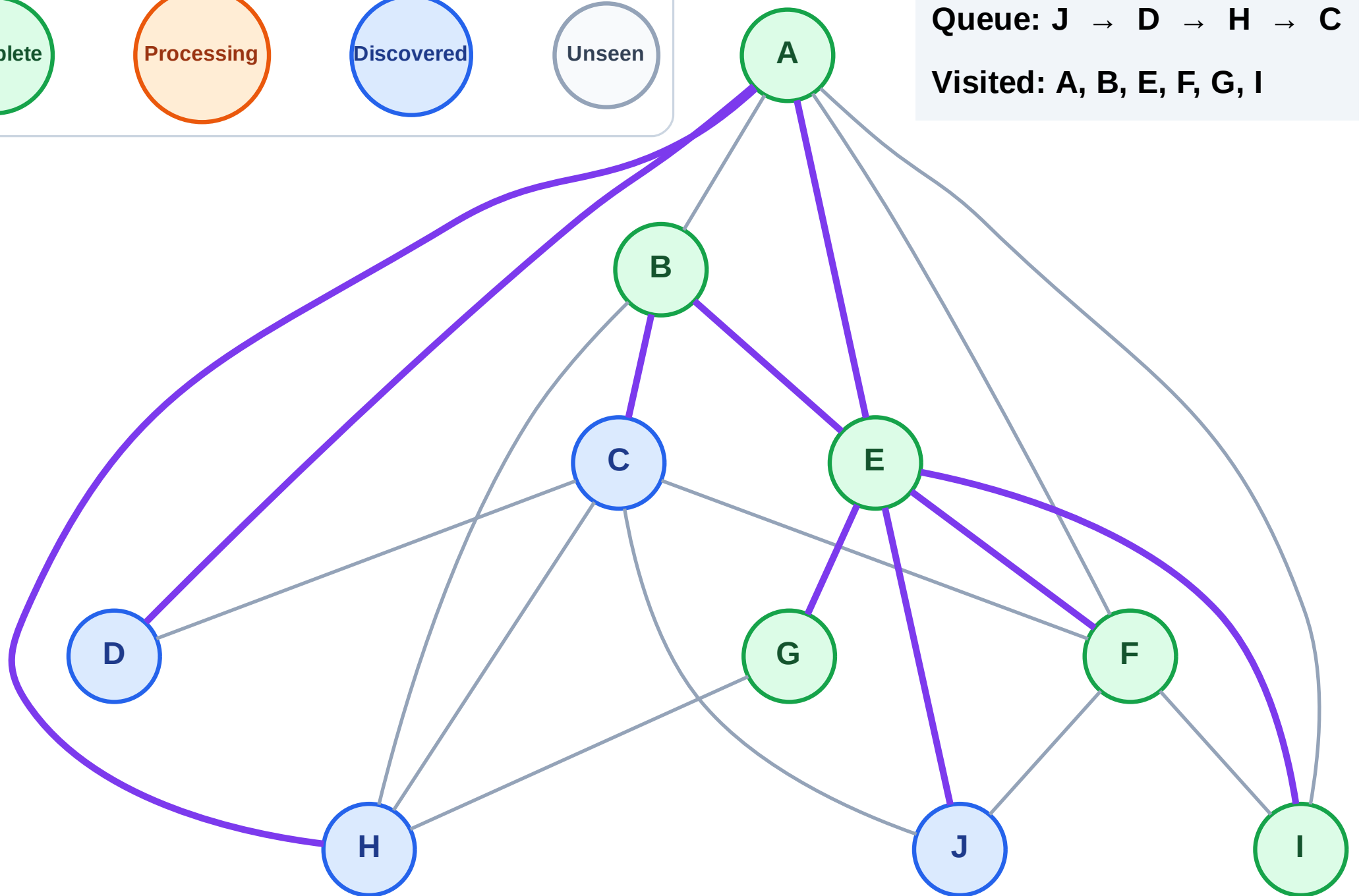
Processing

Discovered

Unseen

Queue: J → D → H → C

Visited: A, B, E, F, G, I



# BFS Traversal

Step 14: Process node J

## Legend

Complete

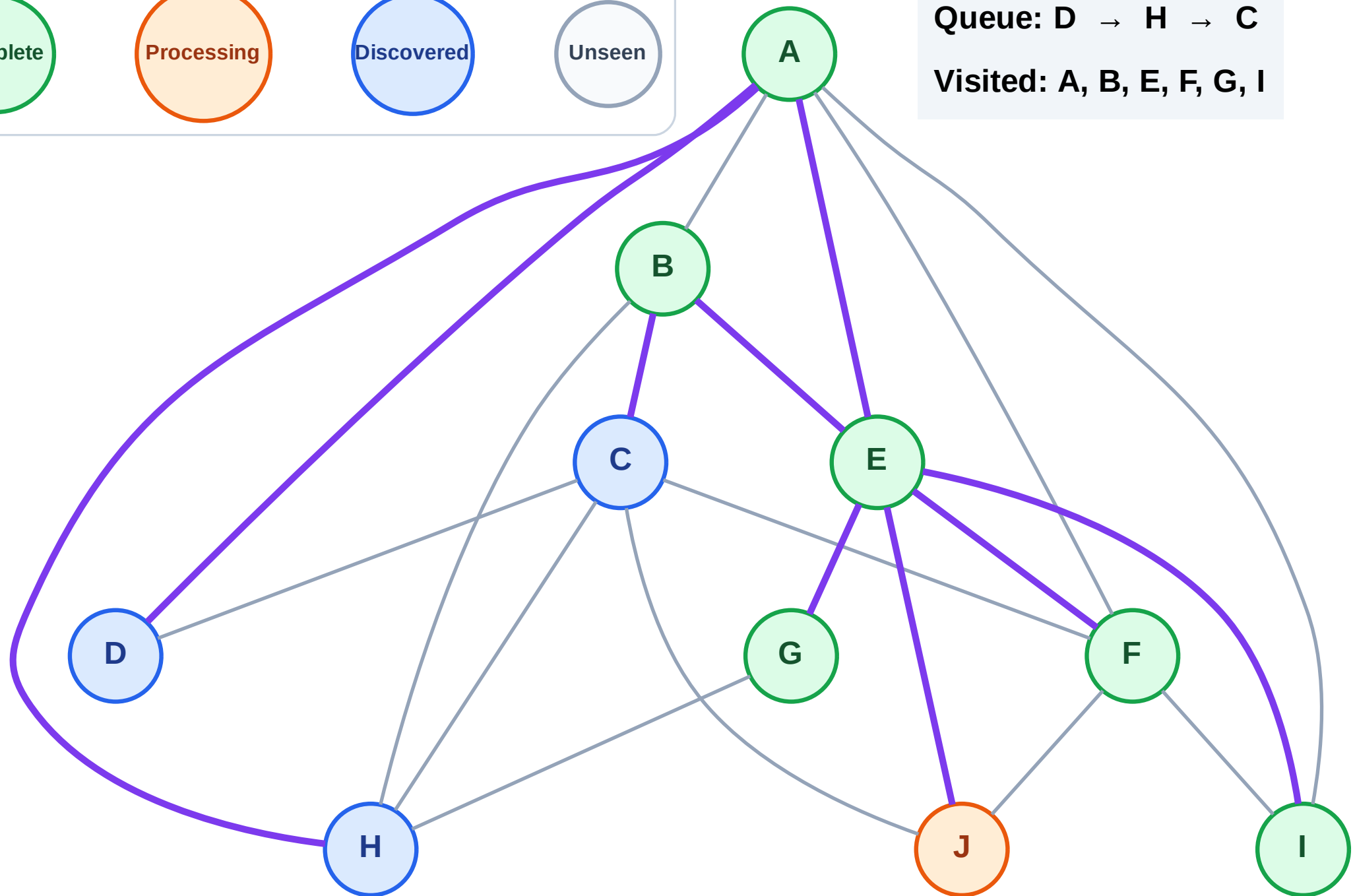
Processing

Discovered

Unseen

Queue: D → H → C

Visited: A, B, E, F, G, I



# BFS Traversal

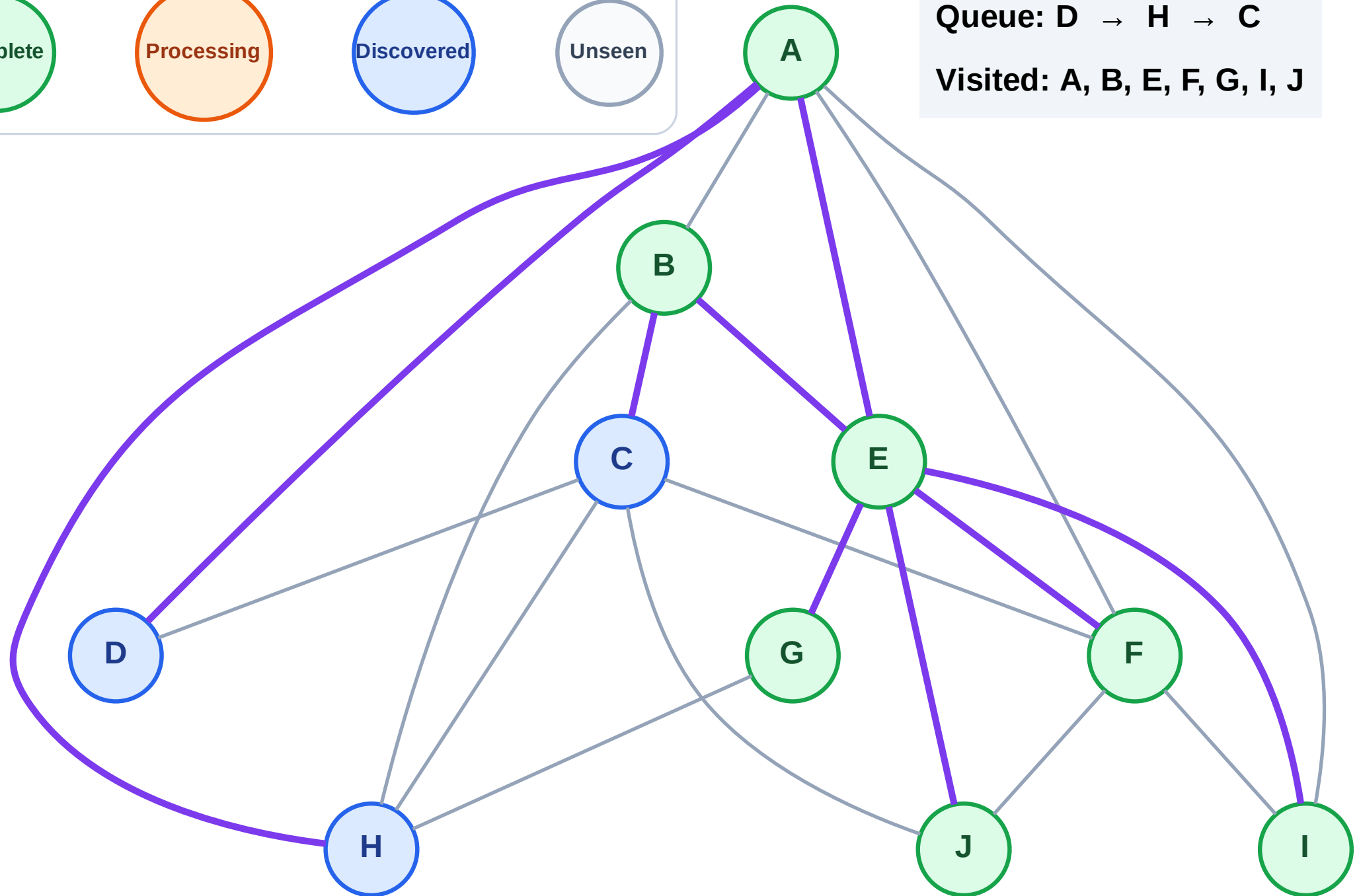
Step 15: No new neighbors from J

## Legend



Queue: D → H → C

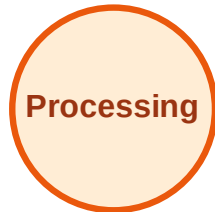
Visited: A, B, E, F, G, I, J



# BFS Traversal

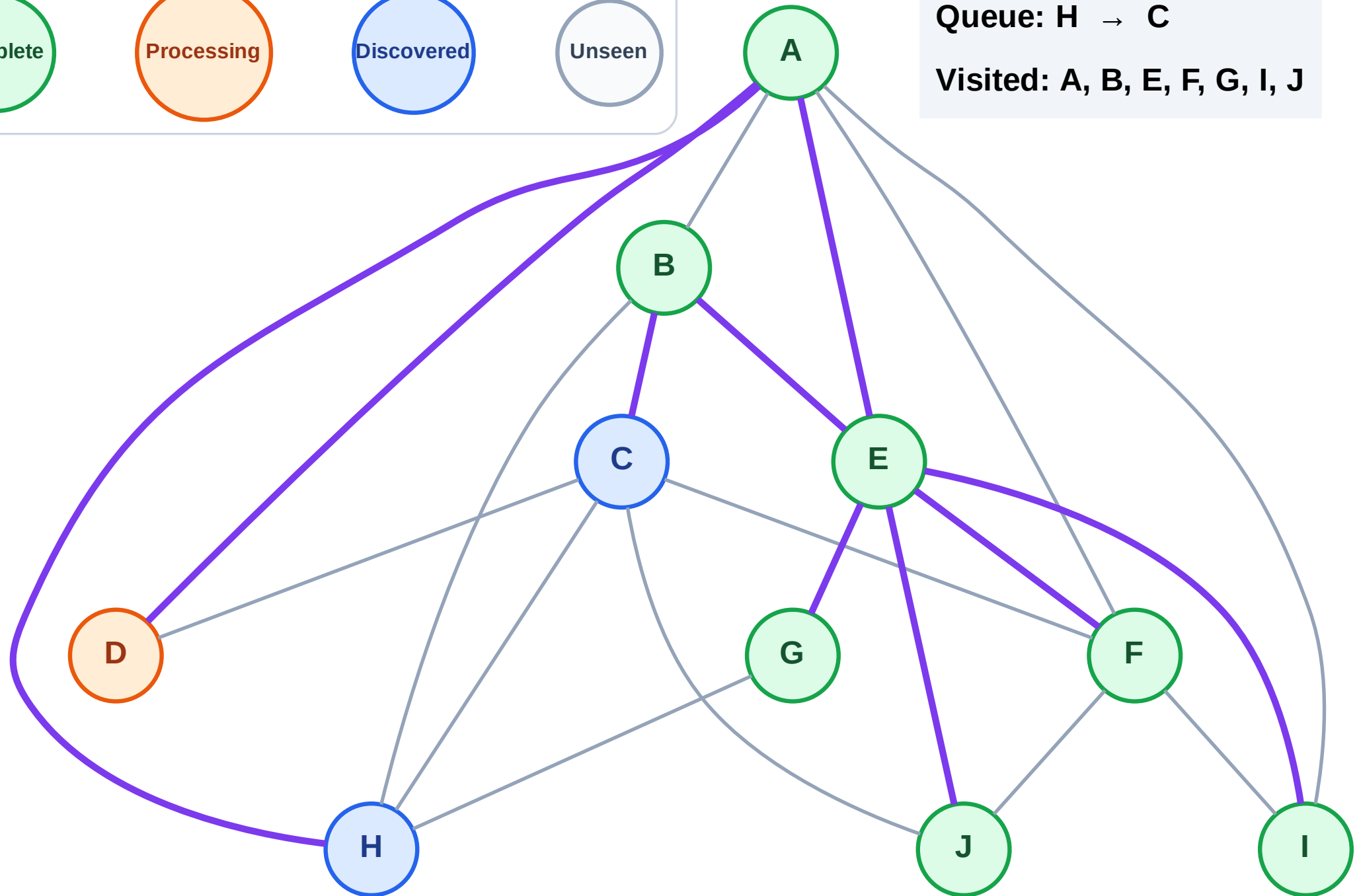
Step 16: Process node D

## Legend



Queue: H → C

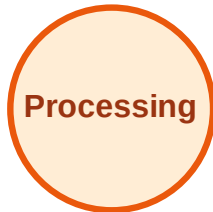
Visited: A, B, E, F, G, I, J



# BFS Traversal

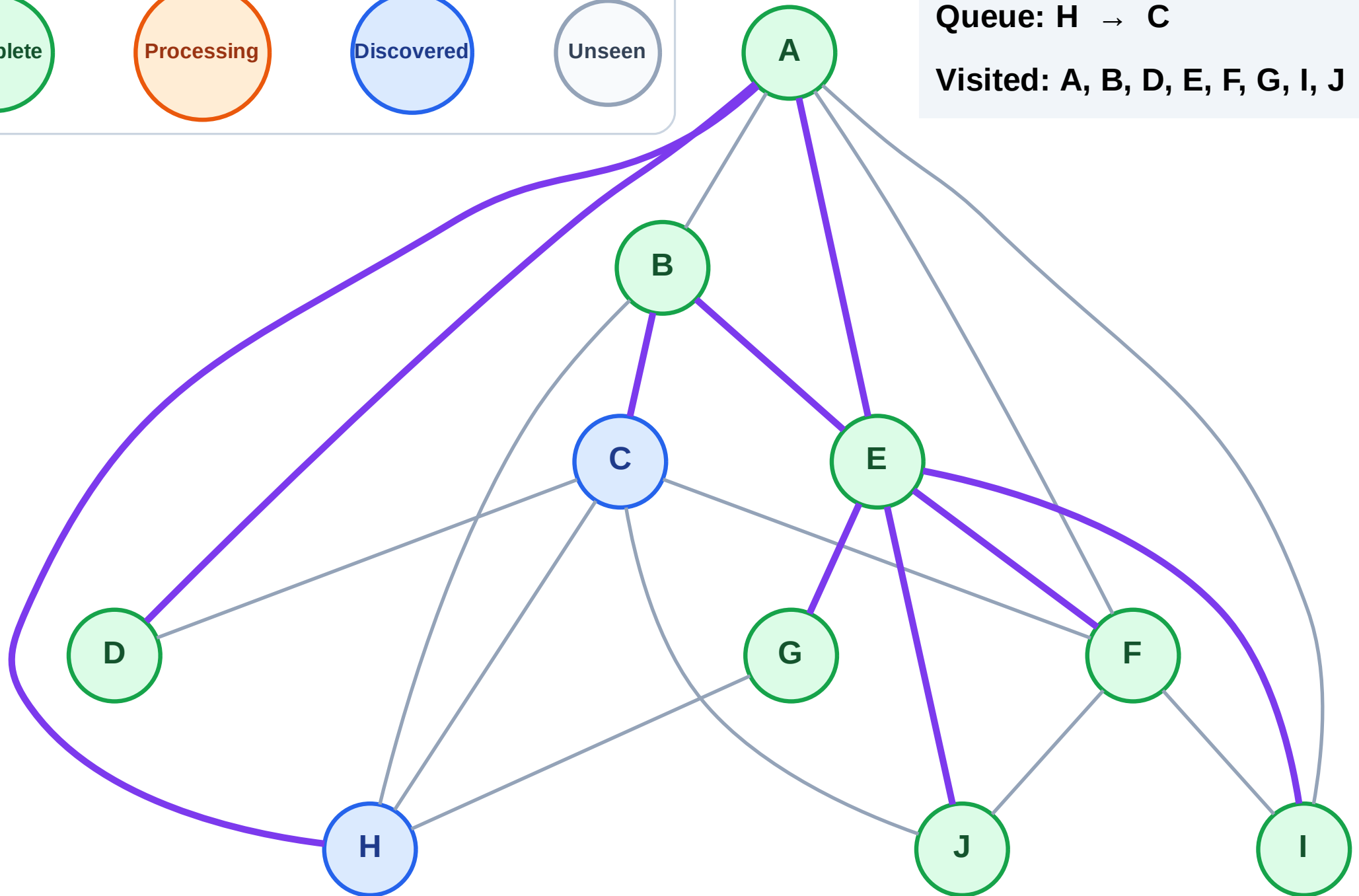
Step 17: No new neighbors from D

## Legend



Queue: H → C

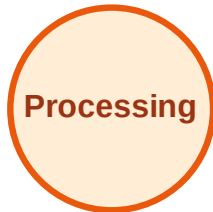
Visited: A, B, D, E, F, G, I, J



# BFS Traversal

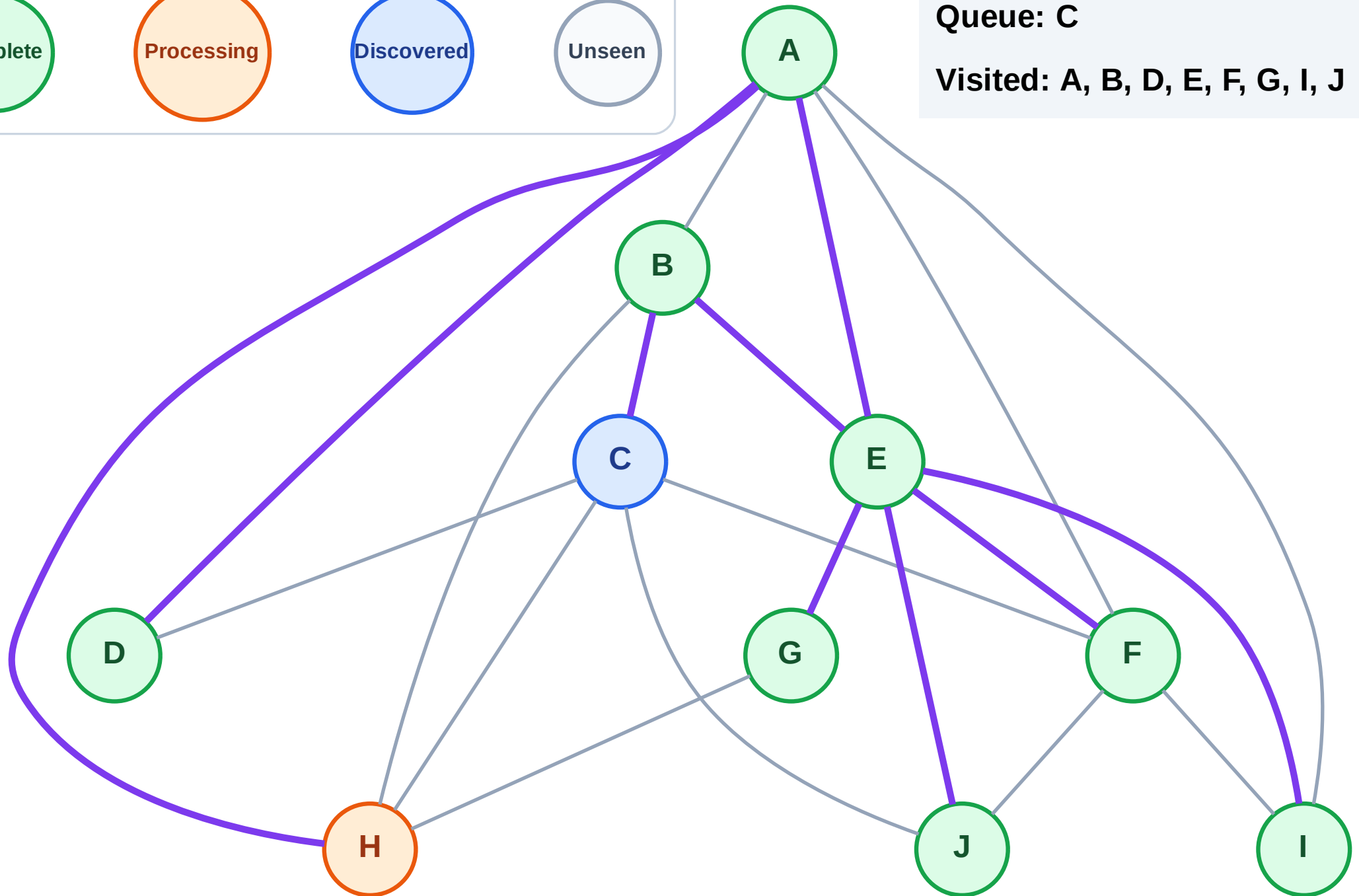
Step 18: Process node H

## Legend



Queue: C

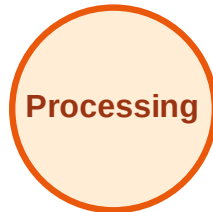
Visited: A, B, D, E, F, G, I, J



# BFS Traversal

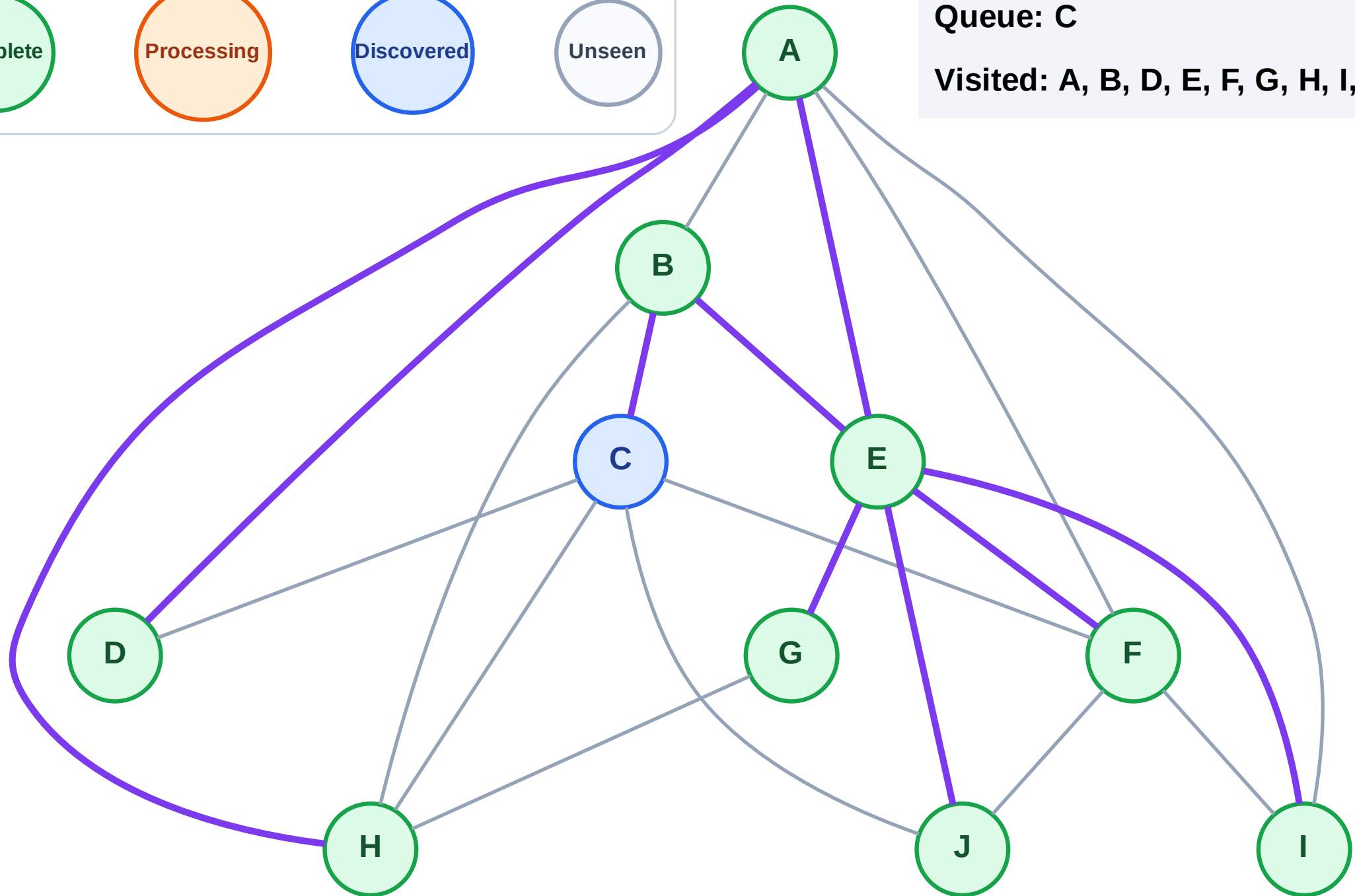
Step 19: No new neighbors from H

## Legend



Queue: C

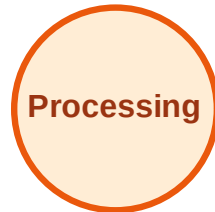
Visited: A, B, D, E, F, G, H, I, J



# BFS Traversal

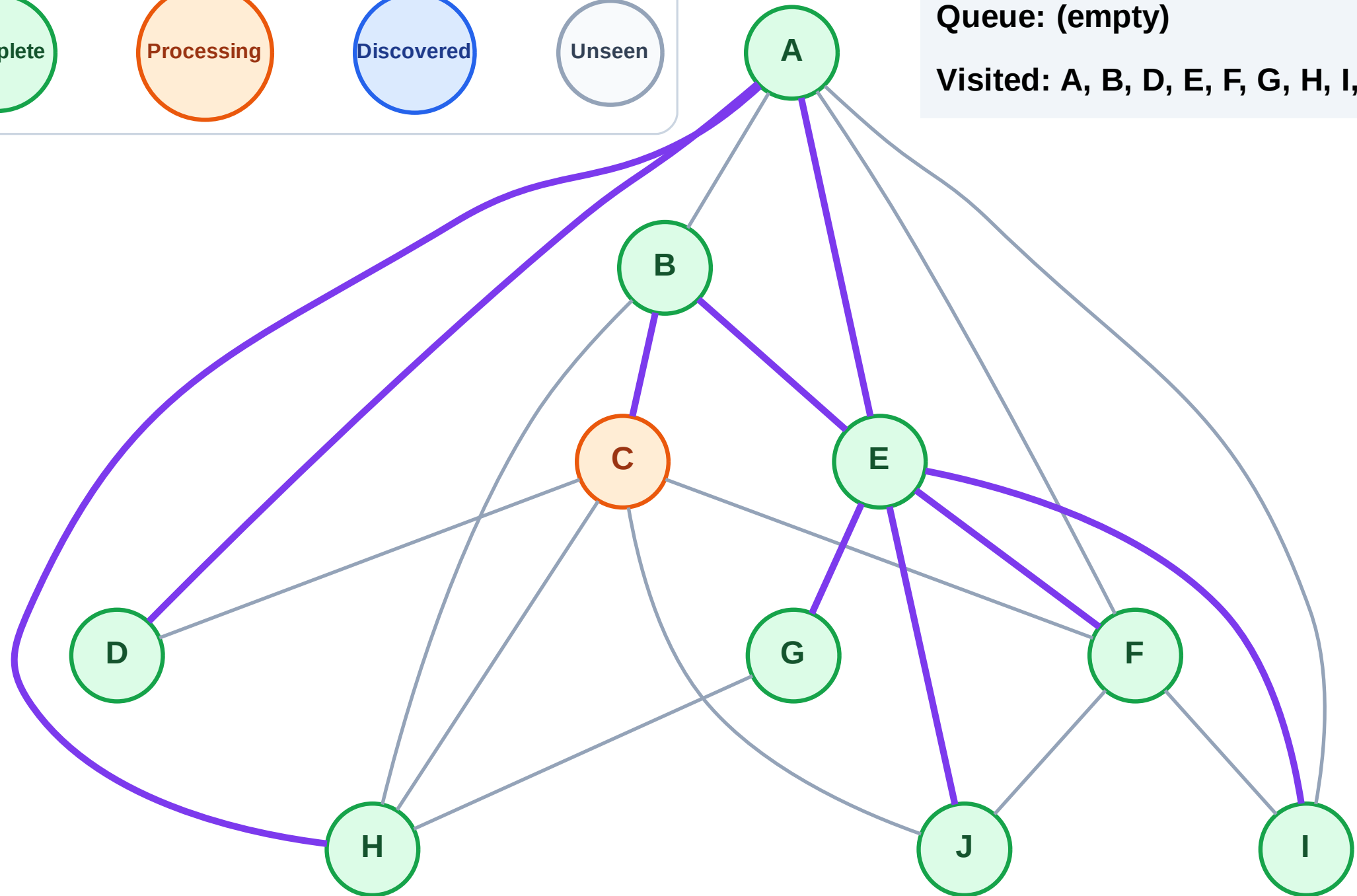
Step 20: Process node C

## Legend



Queue: (empty)

Visited: A, B, D, E, F, G, H, I, J



# BFS Traversal

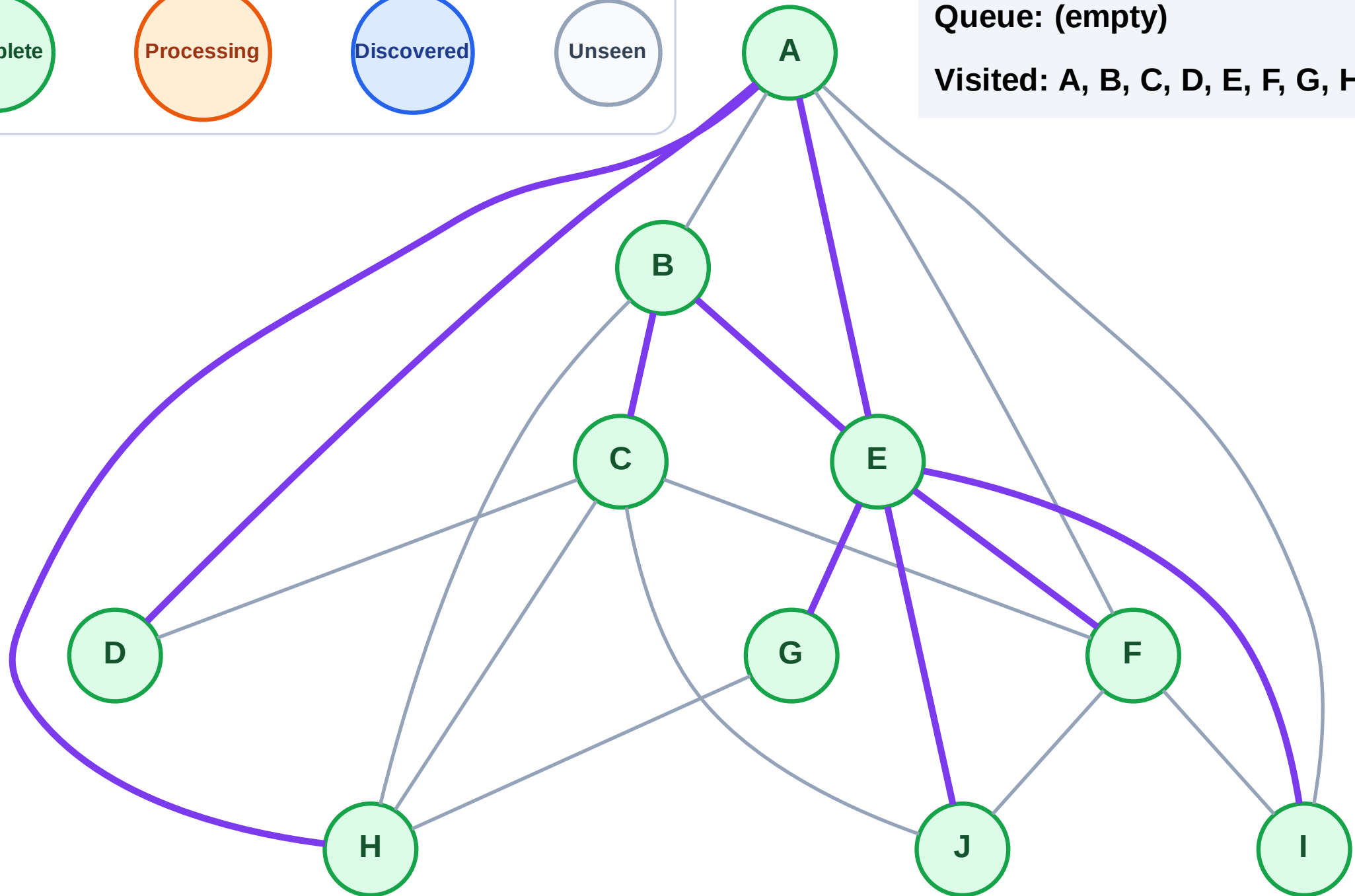
Step 21: No new neighbors from C

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

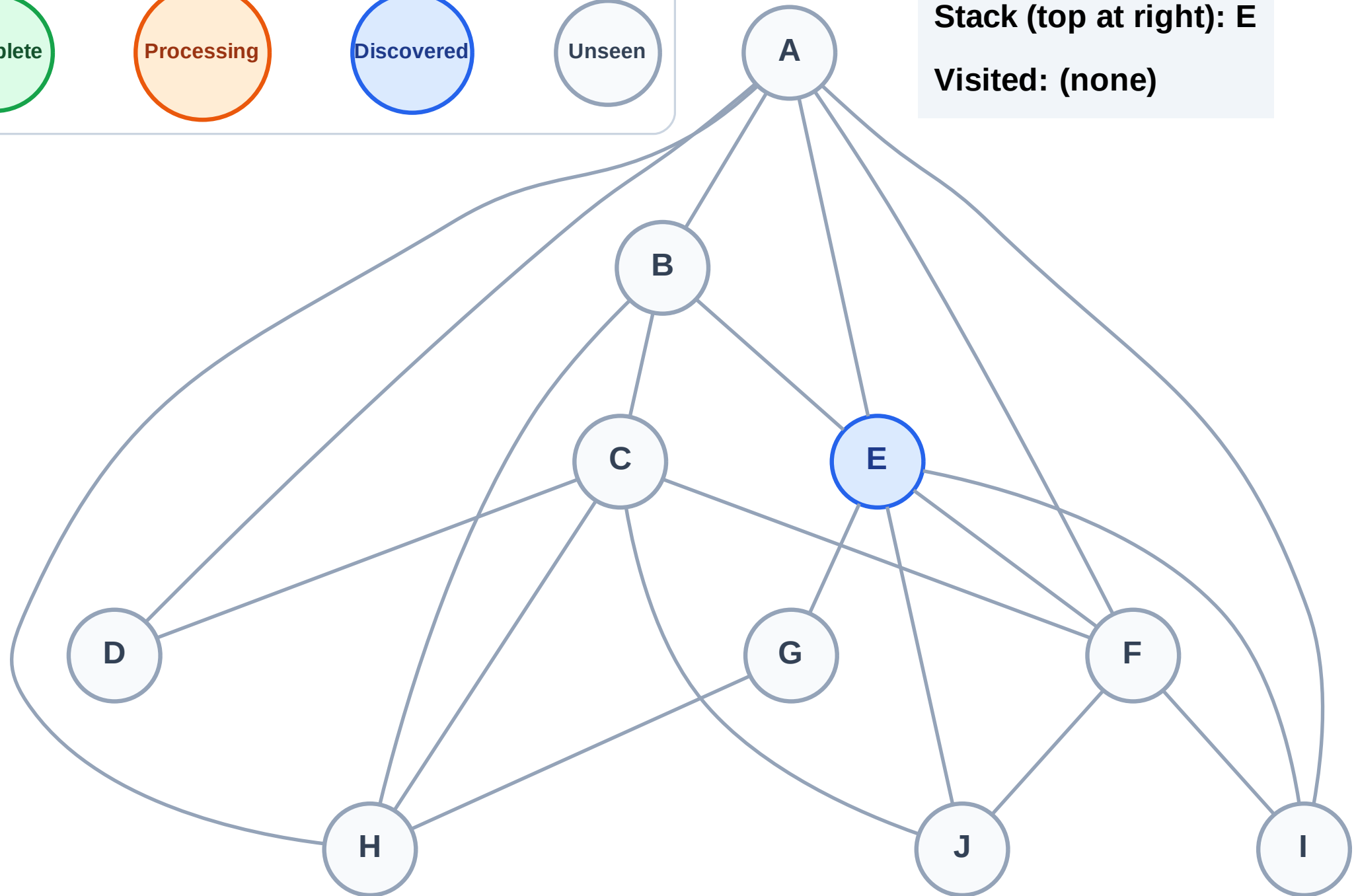
Step 1: Start at E

## Legend



Stack (top at right): E

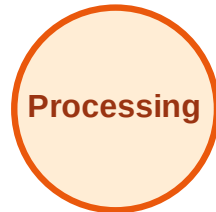
Visited: (none)



# DFS Traversal

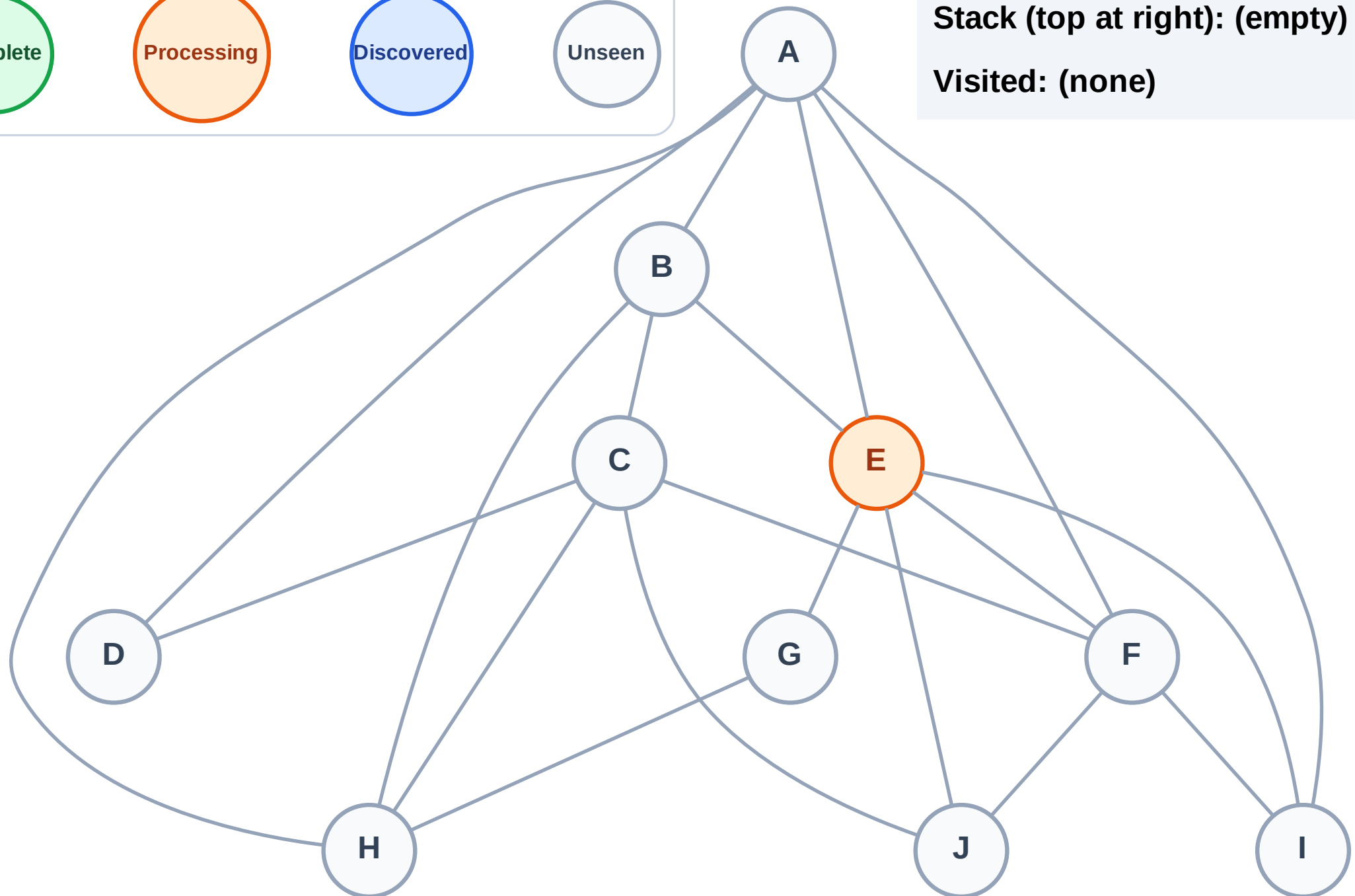
Step 2: Process node E

## Legend



Stack (top at right): (empty)

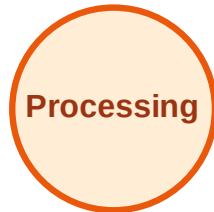
Visited: (none)



# DFS Traversal

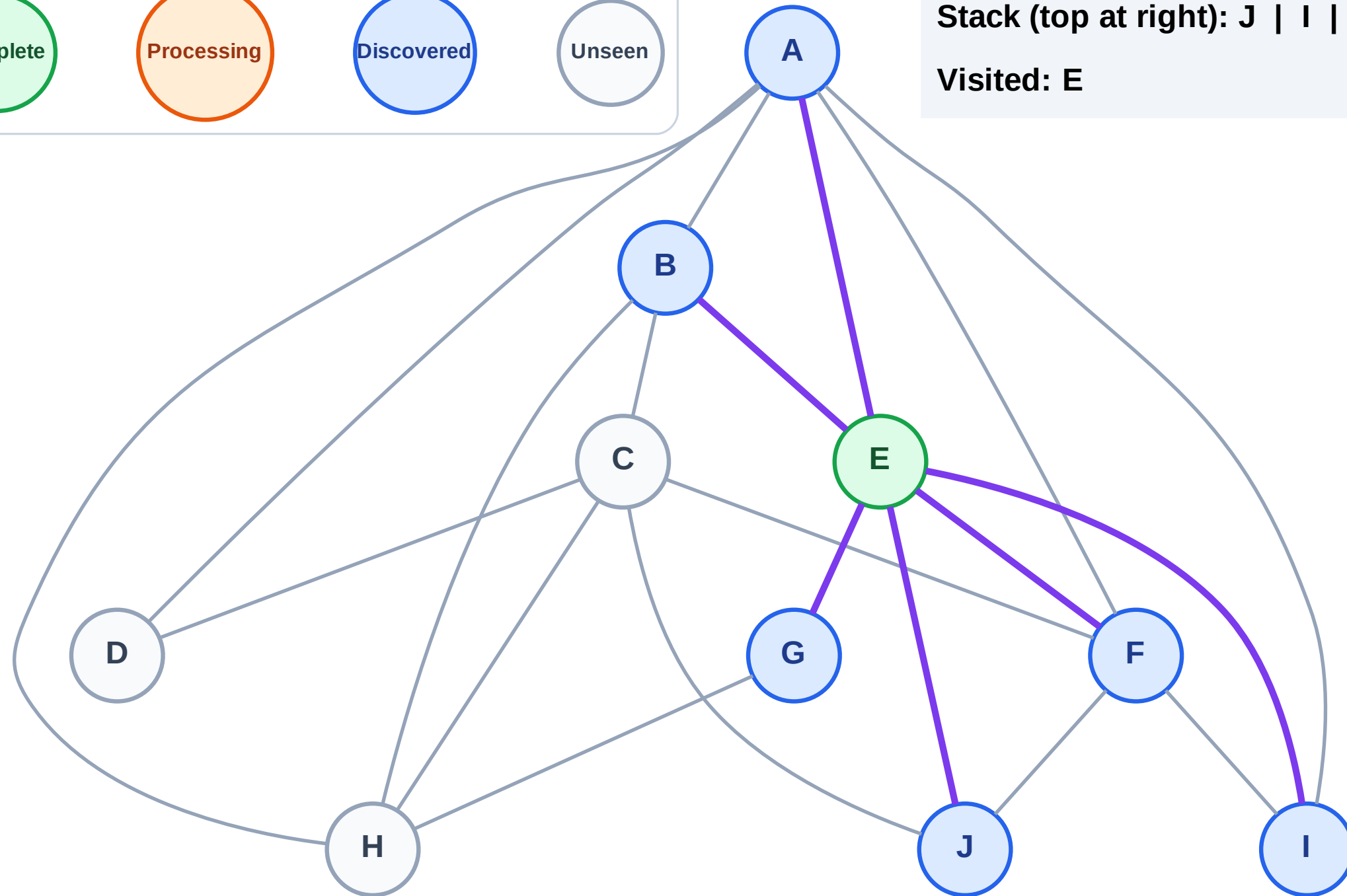
Step 3: Discover J, I, G, F, B, A from E

## Legend



Stack (top at right): J | I | G | F | B | A

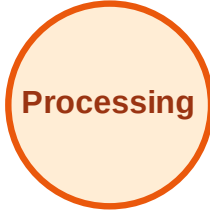
Visited: E



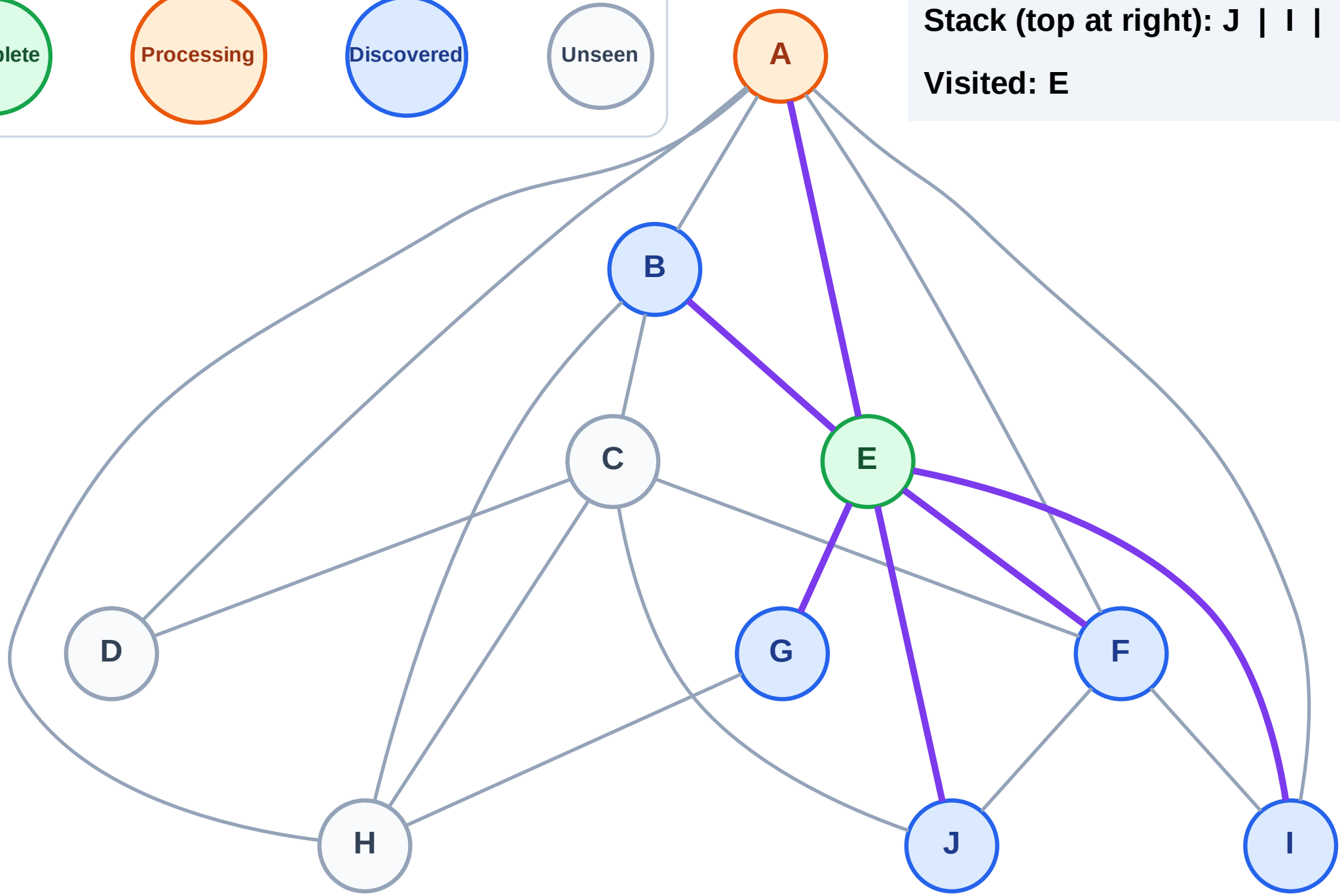
# DFS Traversal

Step 4: Process node A

## Legend



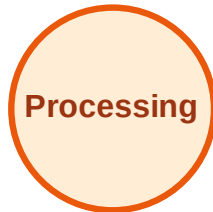
Stack (top at right): J | I | G | F | B  
Visited: E



# DFS Traversal

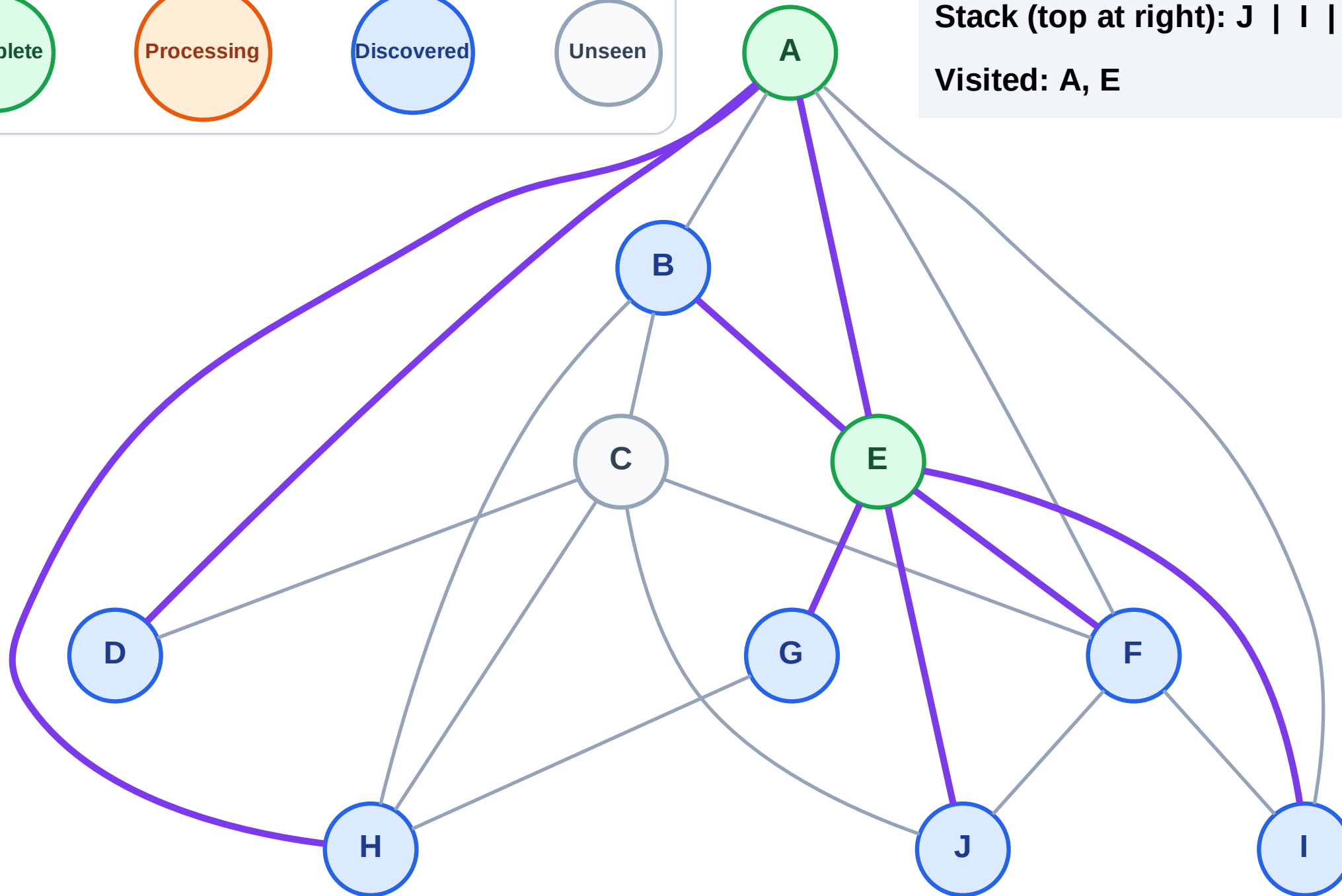
Step 5: Discover H, D from A

## Legend



Stack (top at right): J | I | G | F | B | H | D

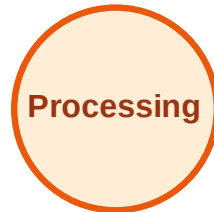
Visited: A, E



# DFS Traversal

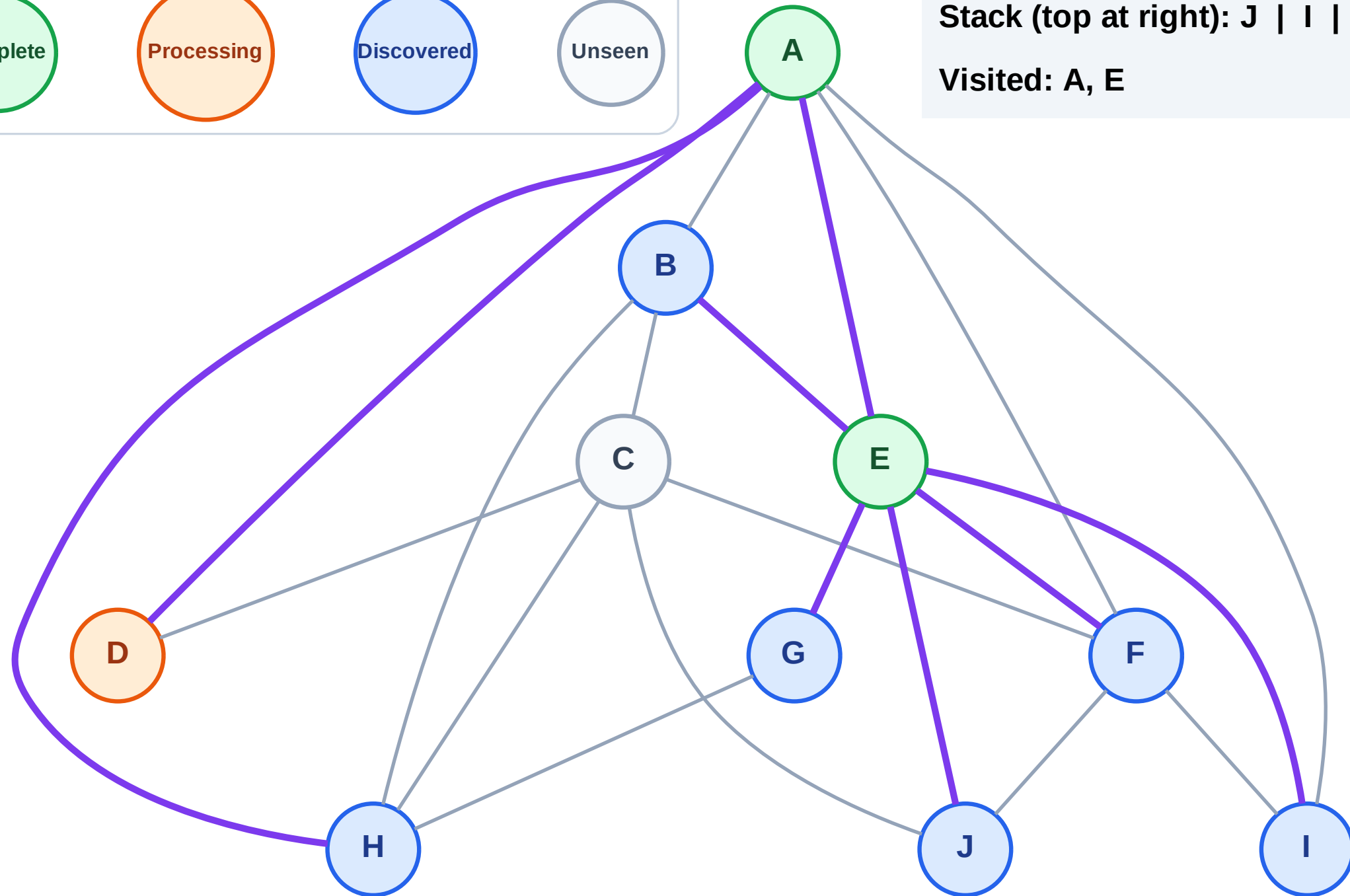
Step 6: Process node D

## Legend



Stack (top at right): J | I | G | F | B | H

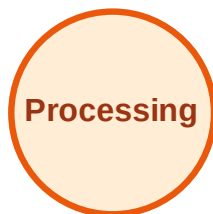
Visited: A, E



# DFS Traversal

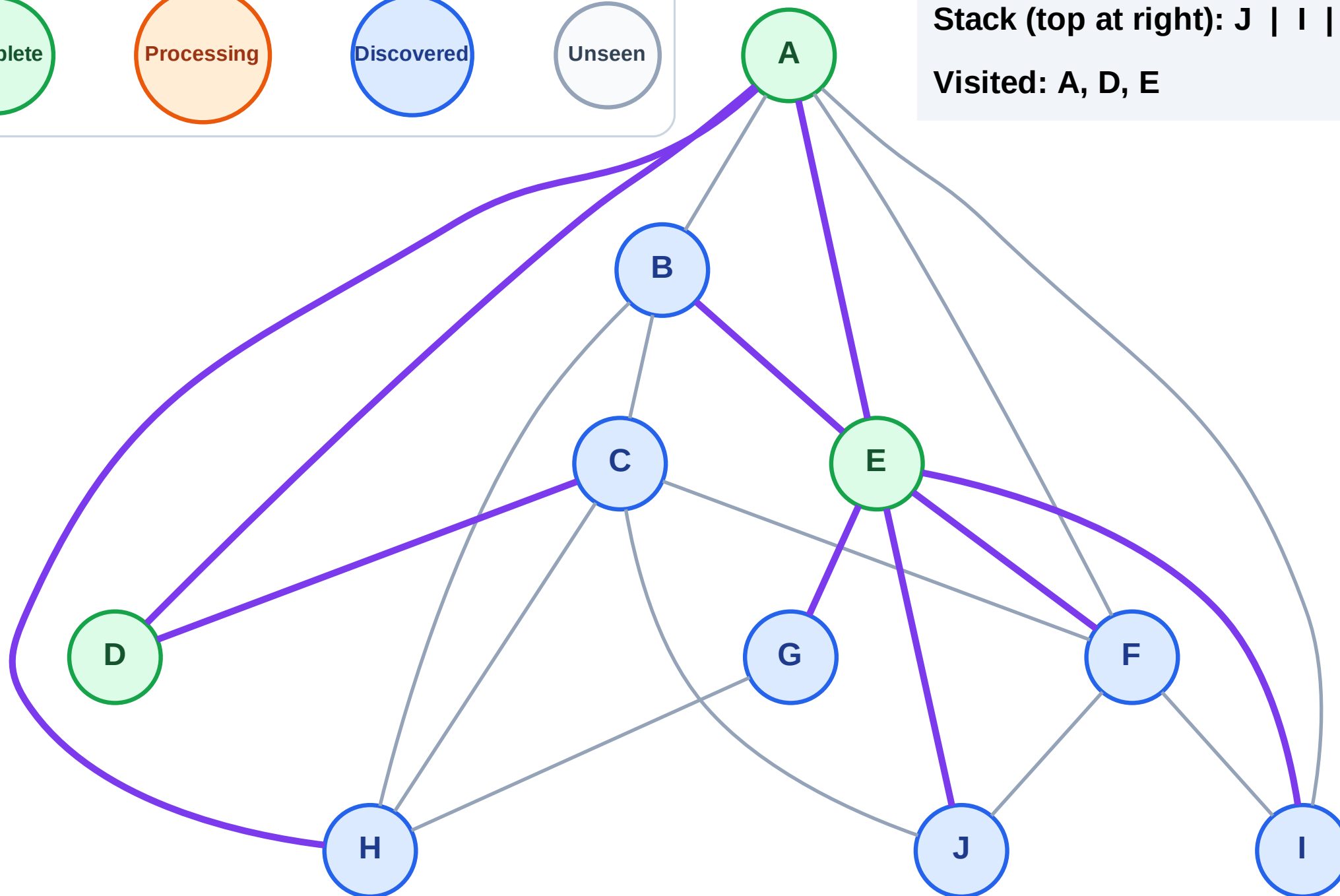
Step 7: Discover C from D

## Legend



Stack (top at right): J | I | G | F | B | H | C

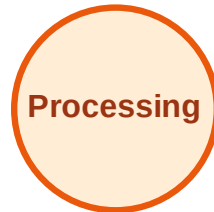
Visited: A, D, E



# DFS Traversal

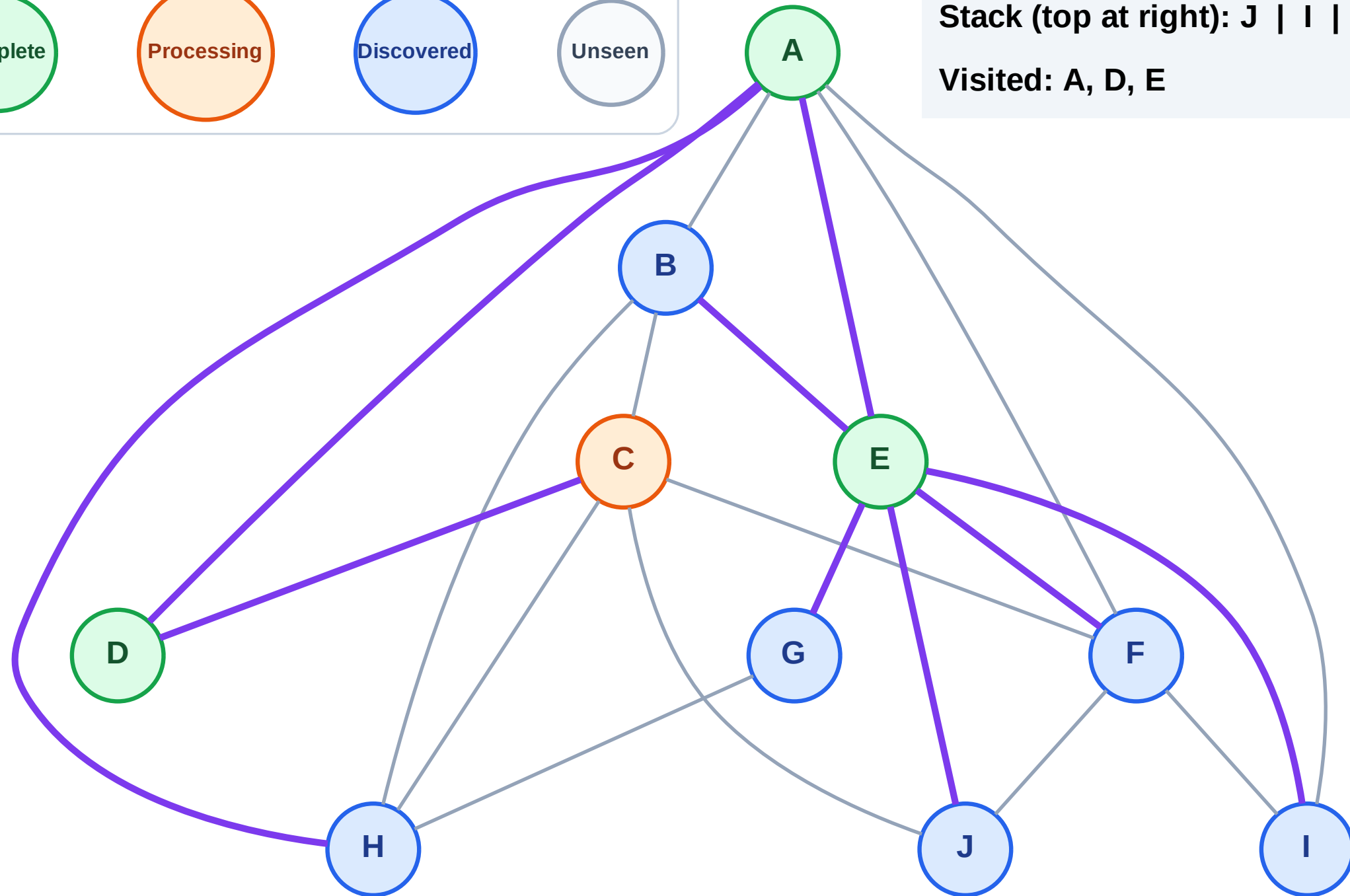
Step 8: Process node C

## Legend



Stack (top at right): J | I | G | F | B | H

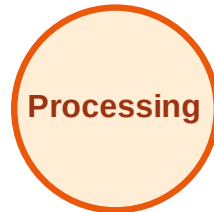
Visited: A, D, E



# DFS Traversal

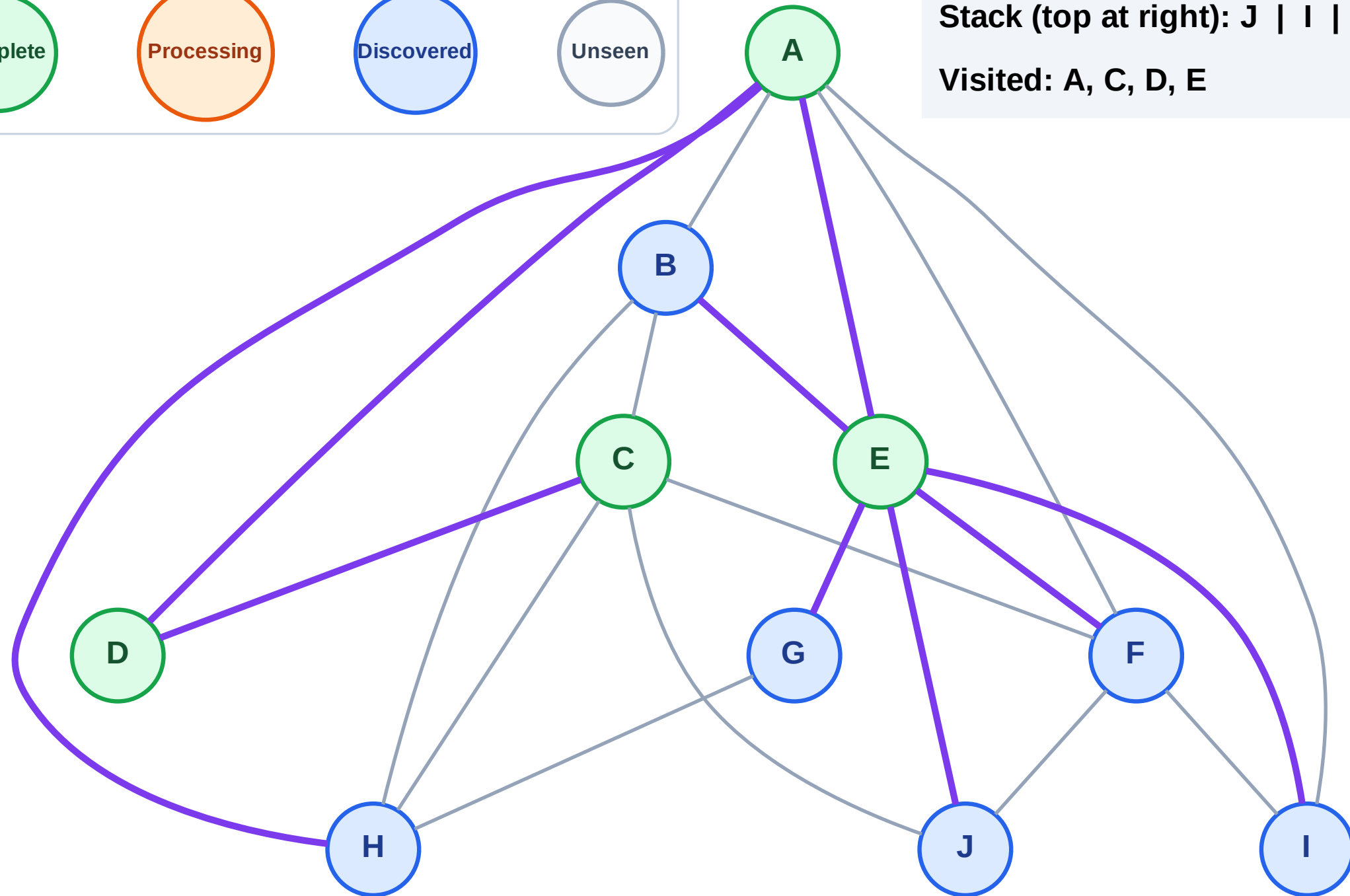
Step 9: No new neighbors from C

## Legend



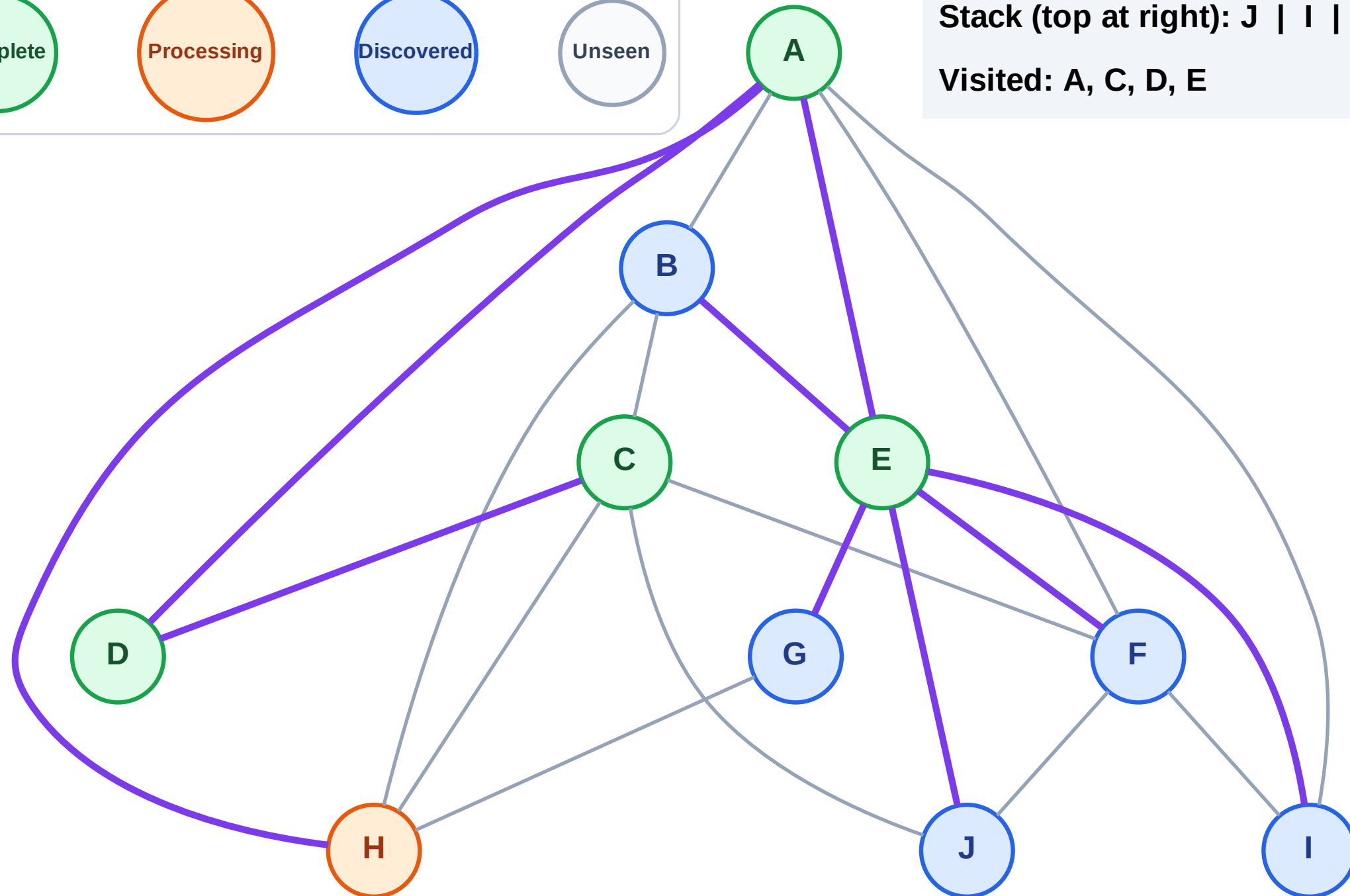
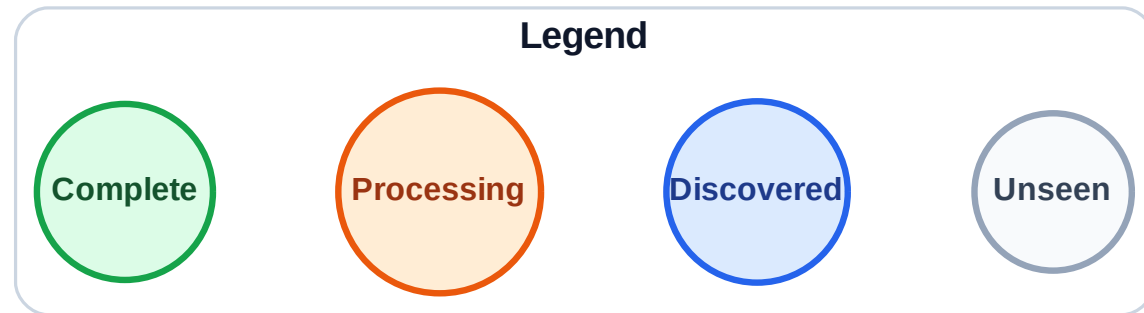
Stack (top at right): J | I | G | F | B | H

Visited: A, C, D, E



### Step 10: Process node H

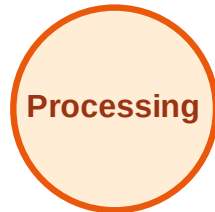
### Step 10: Process node H



# DFS Traversal

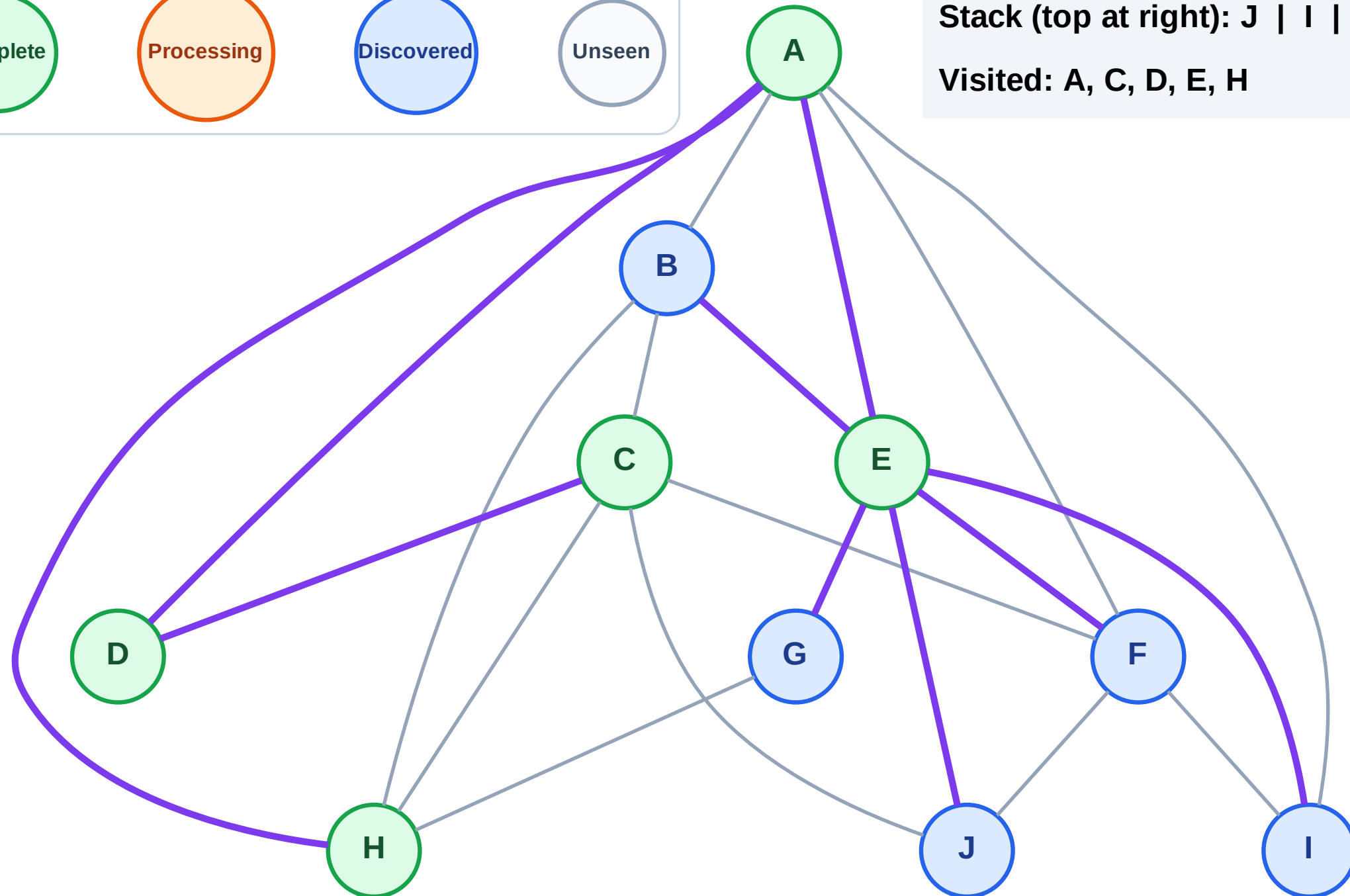
Step 11: No new neighbors from H

## Legend



Stack (top at right): J | I | G | F | B

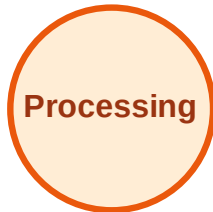
Visited: A, C, D, E, H



# DFS Traversal

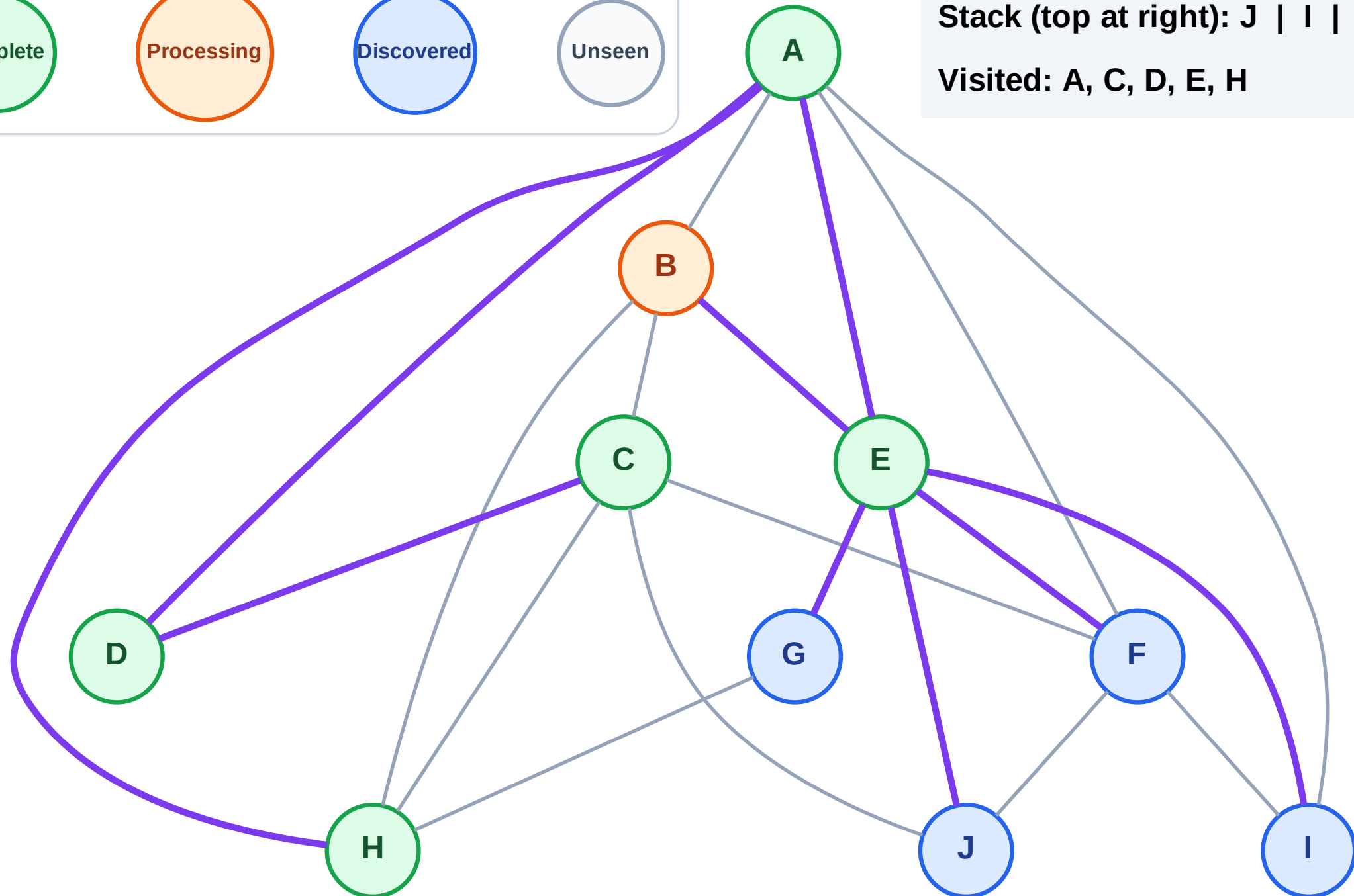
Step 12: Process node B

## Legend



Stack (top at right): J | I | G | F

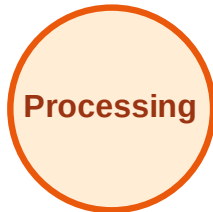
Visited: A, C, D, E, H



# DFS Traversal

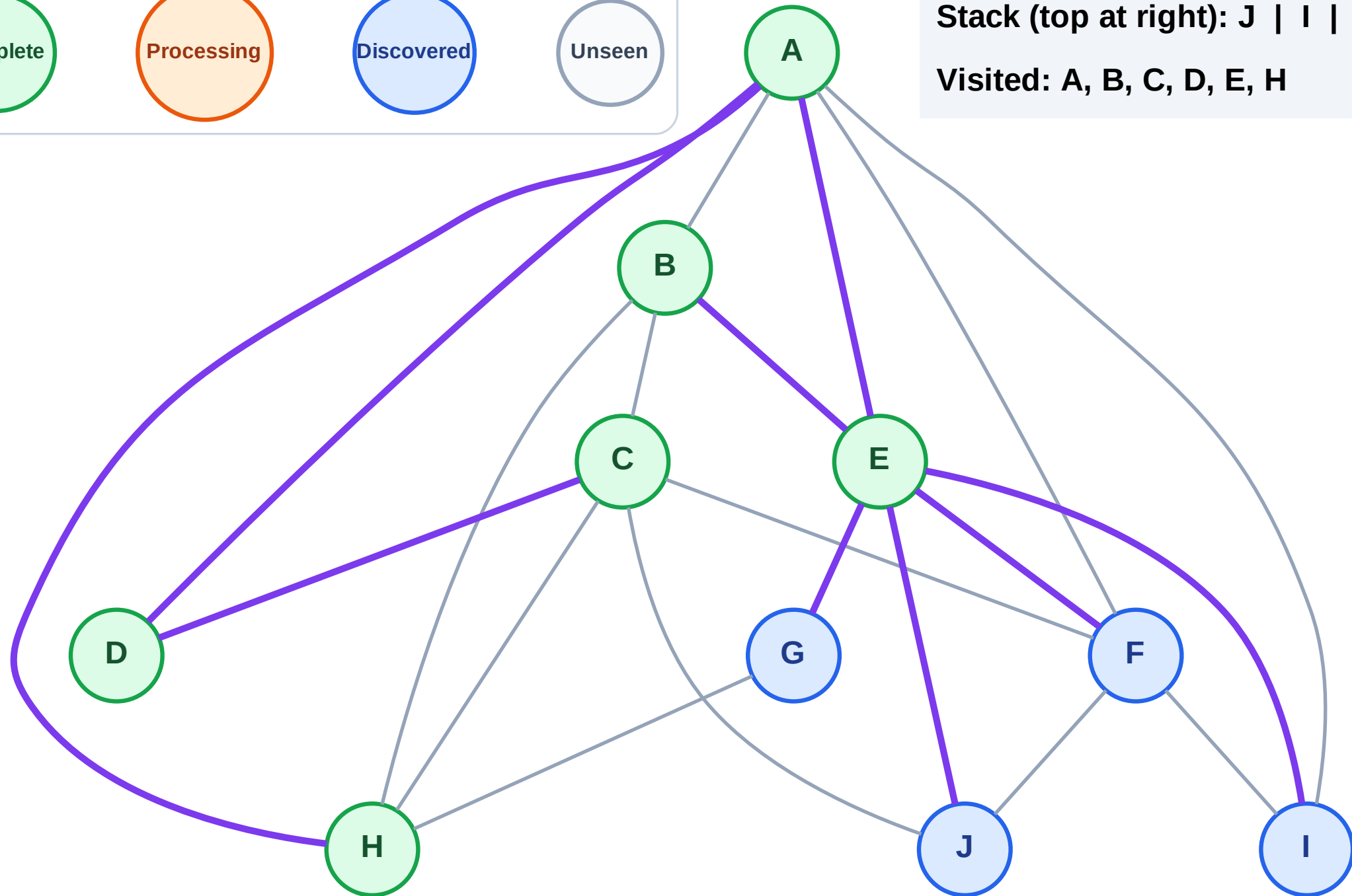
Step 13: No new neighbors from B

## Legend



Stack (top at right): J | I | G | F

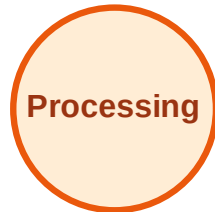
Visited: A, B, C, D, E, H



# DFS Traversal

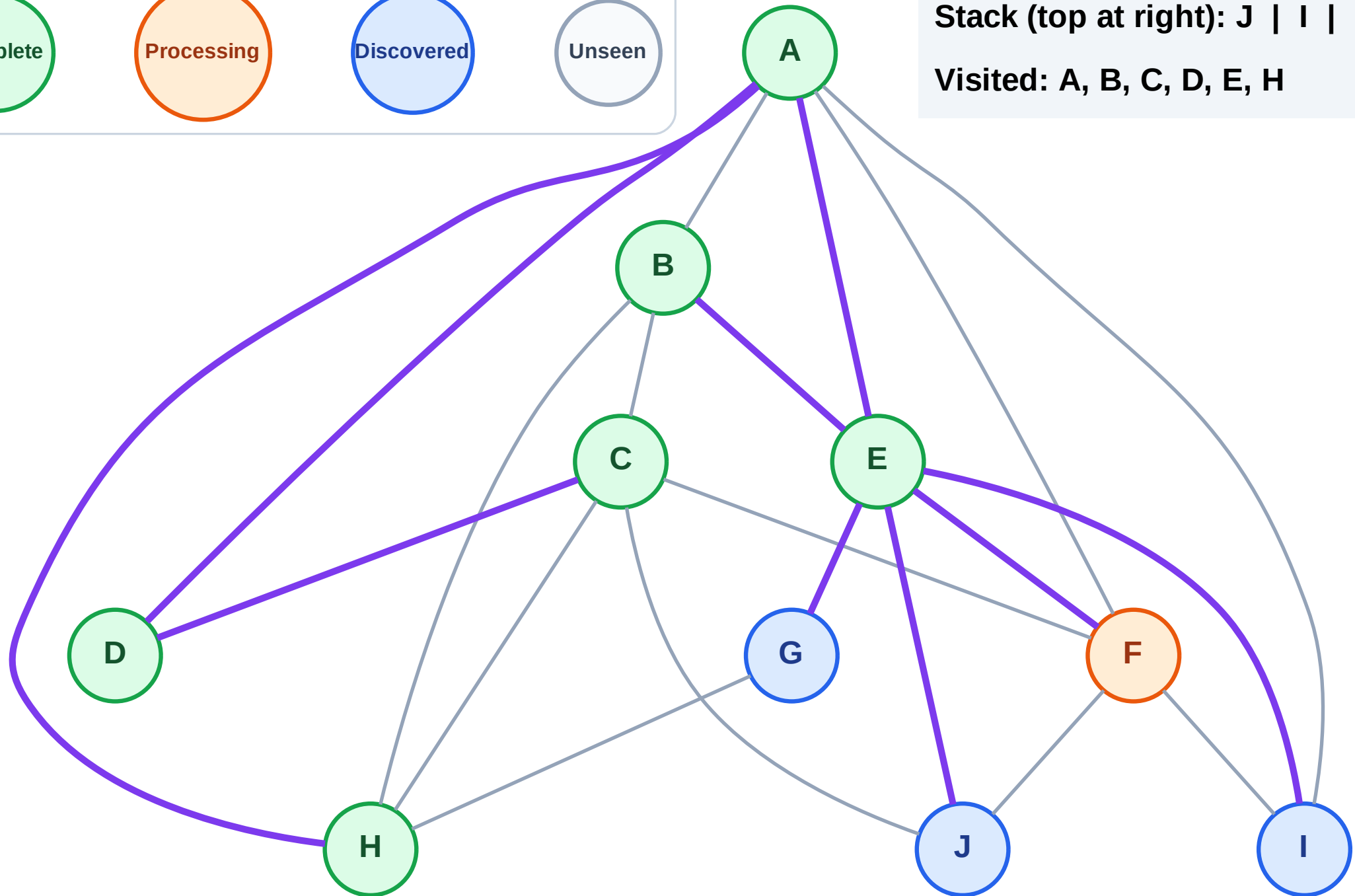
Step 14: Process node F

## Legend



Stack (top at right): J | I | G

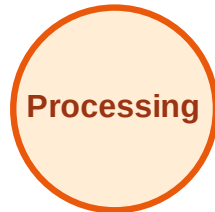
Visited: A, B, C, D, E, H



# DFS Traversal

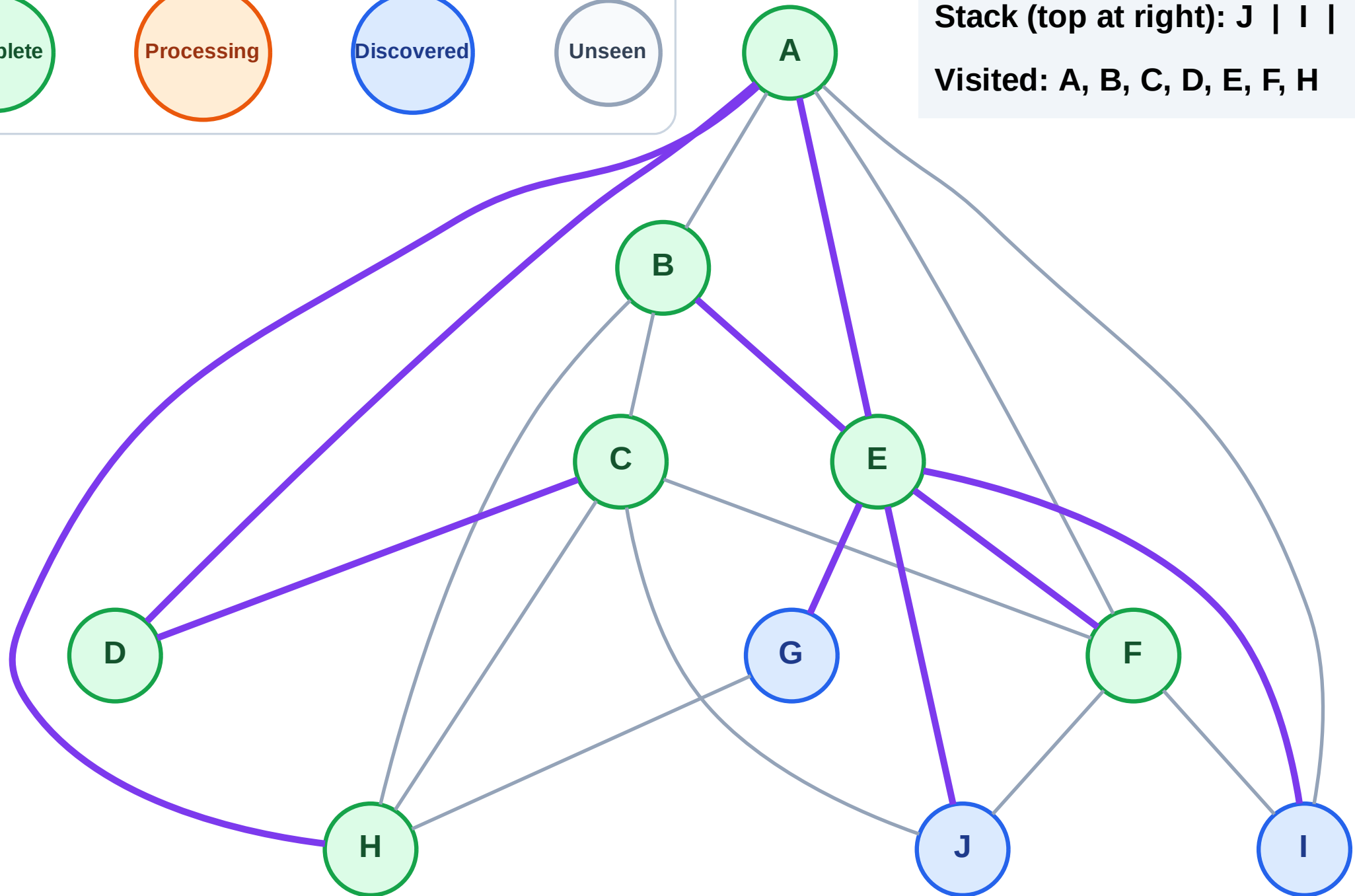
Step 15: No new neighbors from F

## Legend



Stack (top at right): J | I | G

Visited: A, B, C, D, E, F, H



# DFS Traversal

Step 16: Process node G

## Legend

Complete

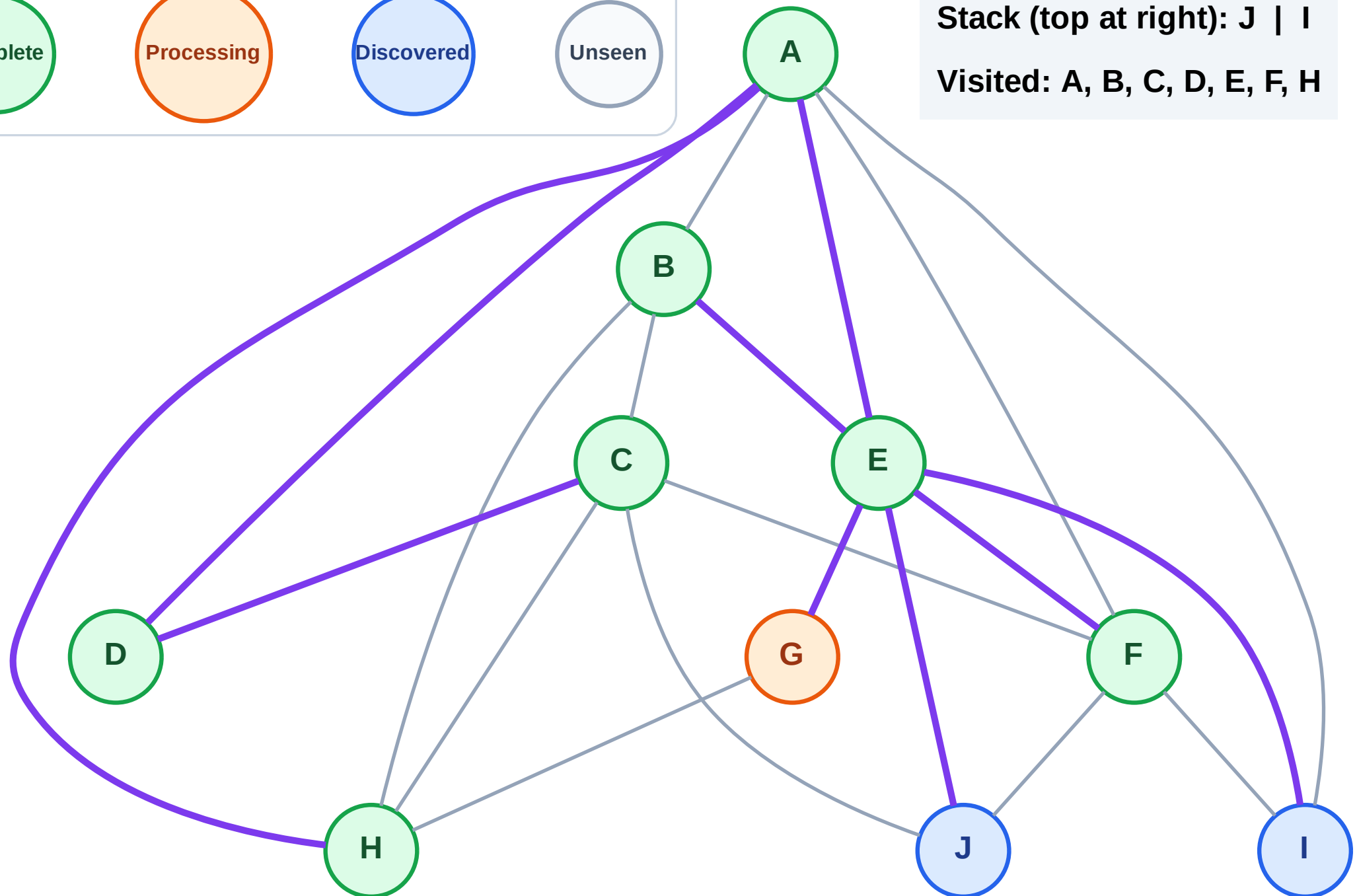
Processing

Discovered

Unseen

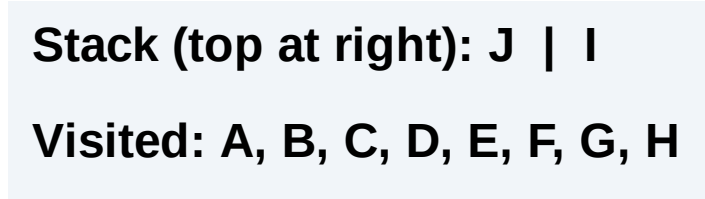
Stack (top at right): J | I

Visited: A, B, C, D, E, F, H



Step 17: No new neighbors from G

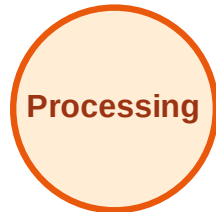
Step 17: No new neighbors from G



# DFS Traversal

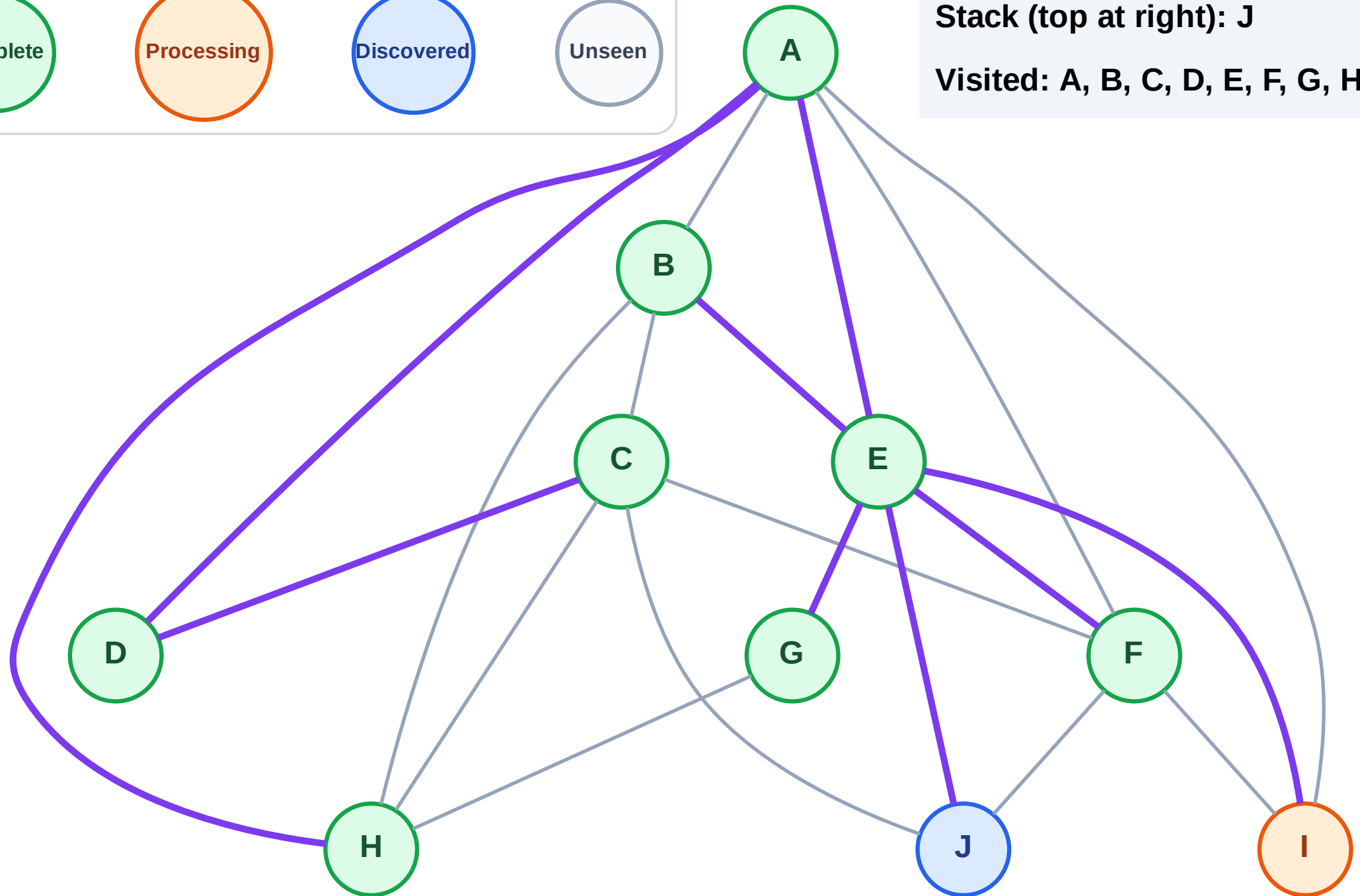
Step 18: Process node I

## Legend



Stack (top at right): J

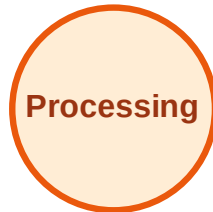
Visited: A, B, C, D, E, F, G, H



# DFS Traversal

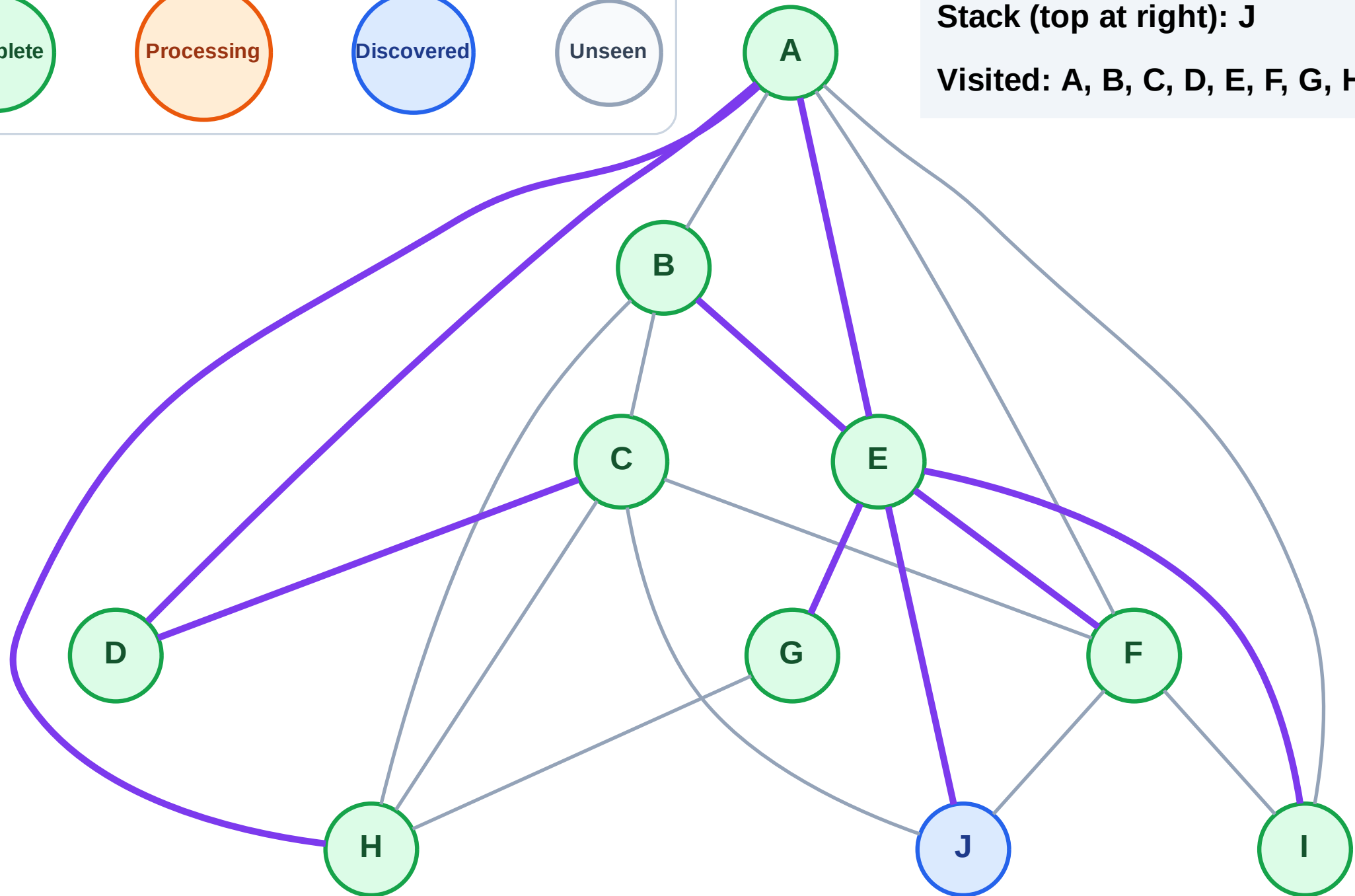
Step 19: No new neighbors from I

## Legend



Stack (top at right): J

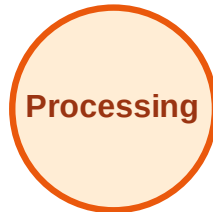
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

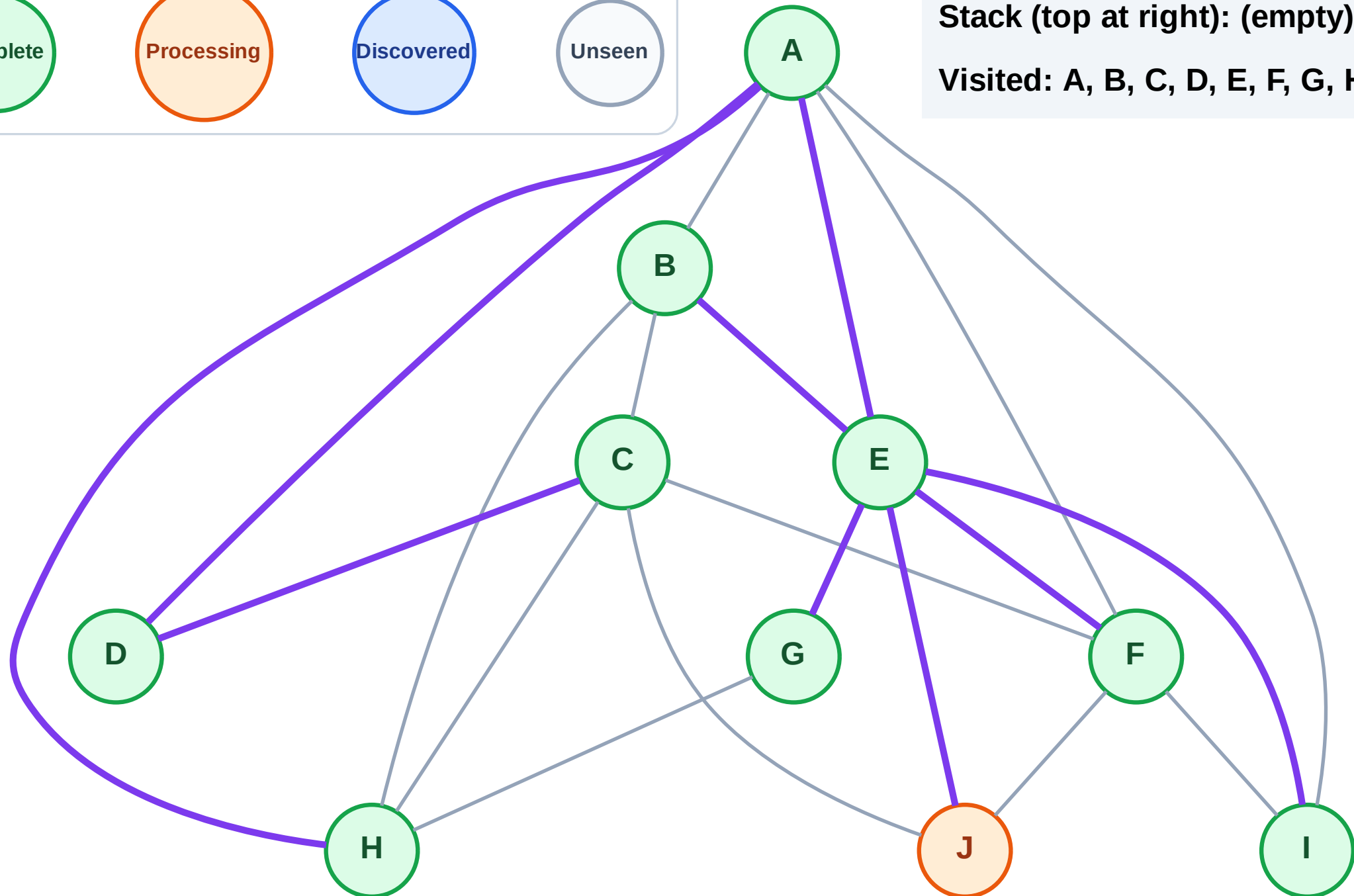
Step 20: Process node J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

